ALGEBRA II · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

84 question sets • 2525 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

CONTENTS

Set 1	**HSN-CN.A.1**	Write complex numbers in the form a + bi	7
Set 2	**HSN-CN.A.2**	Add and subtract complex numbers	44
Set 3	**HSN-CN.A.2**	Multiply complex numbers and simplify powers of i	81
Set 4	**HSN-CN.C.7**	Solve quadratic equations with complex solutions	118
Set 5	**HSN-CN.C.8**	Factor polynomials over the complex numbers	155
Set 6	**HSN-CN.C.9**	Use the Fundamental Theorem of Algebra	192
Set 7	**HSA-SSE.A.1.a**	Interpret the parts of polynomial, rational, and exponential expressions in context	229
Set 8	**HSA-SSE.A.1.b**	Interpret complicated expressions by viewing parts as single entities	266
Set 9	**HSA-SSE.A.2**	Factor higher-degree expressions by recognizing differences of squares and sums	303
Set 10	**HSA-SSE.A.2**	Rewrite expressions in quadratic form and factor	340
Set 11	**HSA-SSE.B.4**	Derive the formula for the sum of a finite geometric series	377
Set 12	**HSA-SSE.B.4**	Use the finite geometric series formula to solve problems	414
Set 13	**HSA-APR.A.1**	Add and subtract polynomials of higher degree	451
Set 14	**HSA-APR.A.1**	Multiply polynomials of higher degree	488
Set 15	**HSA-APR.B.2**	Find remainders with the Remainder Theorem	525
Set 16	**HSA-APR.B.2**	Use p(a) = 0 to find factors	562
Set 17	**HSA-APR.B.3**	Identify zeros and multiplicities from a factorization	599

CONTENTS (continued)

Set 18	**HSA-APR.B.3**	Construct a rough graph of a polynomial from its zeros	636
Set 19	**HSA-APR.C.4**	Prove polynomial identities	679
Set 20	**HSA-APR.C.4**	Use polynomial identities to describe numerical relationships	716
Set 21	**HSA-APR.C.5**	Expand with the Binomial Theorem	753
Set 22	**HSA-APR.D.6**	Rewrite rational expressions by polynomial division	790
Set 23	**HSA-APR.D.6**	Rewrite simple rational expressions by inspection	827
Set 24	**HSA-APR.D.7**	Multiply and divide rational expressions	864
Set 25	**HSA-APR.D.7**	Add and subtract rational expressions	901
Set 26	**HSA-CED.A.1**	Create and solve rational, radical, and exponential equations from contexts	938
Set 27	**HSA-CED.A.1**	Create and solve nonlinear inequalities from contexts	975
Set 28	**HSA-CED.A.2**	Create and graph nonlinear two-variable models	1012
Set 29	**HSA-CED.A.3**	Represent nonlinear constraints and judge viable options	1049
Set 30	**HSA-CED.A.4**	Rearrange rational, radical, and exponential formulas	1086
Set 31	**HSA-REI.A.2**	Solve rational equations and identify extraneous solutions	1123
Set 32	**HSA-REI.A.2**	Solve radical equations and identify extraneous solutions	1160
Set 33	**HSA-REI.D.11**	Explain why intersections of nonlinear graphs solve f(x) = g(x)	1197
Set 34	**HSA-REI.D.11**	Approximate solutions of f(x) = g(x)	1234

CONTENTS (continued)

Set 35	**HSF-TF.A.1**	Use radian measure as arc length on the unit circle	1271
Set 36	**HSF-TF.A.2**	Find exact trigonometric values beyond the first quadrant	1308
Set 37	**HSF-TF.A.2**	Explain the extension of sine and cosine to all real numbers	1345
Set 38	**HSF-TF.B.5**	Write sinusoidal models with given amplitude, period, and midline	1382
Set 39	**HSF-TF.B.5**	Interpret the parameters of sinusoidal models	1419
Set 40	**HSF-TF.C.8**	Prove the Pythagorean identity	1456
Set 41	**HSF-TF.C.8**	Use the Pythagorean identity to find trigonometric values	1493
Set 42	**HSF-IF.B.4**	Interpret key features of polynomial and rational models	1530
Set 43	**HSF-IF.B.4**	Interpret periodic models from graphs and tables	1567
Set 44	**HSF-IF.B.5**	State domains of radical, rational, logarithmic, and polynomial models	1604
Set 45	**HSF-IF.B.6**	Calculate and interpret average rates of change of nonlinear models	1641
Set 46	**HSF-IF.C.7.b**	Graph square root and cube root functions with transformations	1678
Set 47	**HSF-IF.C.7.b**	Graph piecewise-defined and absolute value functions	1715
Set 48	**HSF-IF.C.7.c**	Graph polynomial functions showing zeros and end behavior	1752
Set 49	**HSF-IF.C.7.e**	Graph exponential functions with intercepts, asymptote, and end behavior	1789
Set 50	**HSF-IF.C.7.e**	Graph logarithmic functions with intercept, asymptote, and end behavior	1826
Set 51	**HSF-IF.C.7.e**	Graph trigonometric functions showing period, midline, and amplitude	1863

CONTENTS (continued)

Set 52	**HSF-IF.C.8.b**	Rewrite exponential functions to reveal rates over other time units	1900
Set 53	**HSF-IF.C.9**	Compare properties of Algebra II functions in different representations	1937
Set 54	**HSF-BF.A.1.b**	Build models by combining Algebra II function types	1974
Set 55	**HSF-BF.B.3**	Identify vertical shifts, vertical stretches, and horizontal shifts of Algebra II functio	2011
Set 56	**HSF-BF.B.3**	Horizontal stretches, compressions, and reflections f(kx)	2048
Set 57	**HSF-BF.B.3**	Recognize even and odd functions	2085
Set 58	**HSF-BF.B.4.a**	Write inverses of radical, rational, cubic, exponential, and logarithmic functions	2122
Set 59	**HSF-LE.A.4**	Write exact logarithmic solutions of exponential equations	2159
Set 60	**HSF-LE.A.4**	Evaluate logarithmic solutions of exponential models with technology	2196
Set 61	**HSS-ID.A.4**	Estimate population percentages with a normal distribution	2233
Set 62	**HSS-ID.A.4**	Recognize data sets for which a normal model is not appropriate	2270
Set 63	**HSS-IC.A.1**	Distinguish population, sample, parameter, and statistic	2307
Set 64	**HSS-IC.A.2**	Decide whether a model is consistent with observed results using a simulation	2344
Set 65	**HSS-IC.B.3**	Identify sample surveys, experiments, and observational studies	2381
Set 66	**HSS-IC.B.3**	Explain random sampling versus random assignment	2418
Set 67	**HSS-IC.B.4**	Estimate a population proportion with a margin of error from a simulation	2455
Set 68	**HSS-IC.B.4**	Estimate a population mean with a margin of error from a simulation	2492

CONTENTS (continued)

Set 69	**HSS-IC.B.5**	Compare two treatments with a re-randomization simulation	2529
Set 70	**HSS-IC.B.6**	Evaluate claims in reports based on data	2566
Set 71	**HSS-MD.B.6**	Design and judge fair decision procedures with probabilities	2603
Set 72	**HSS-MD.B.7**	Analyze decisions and strategies with probabilities	2640
Set 73	**HSN-CN.A.3**	Use conjugates to find moduli and quotients of complex numbers	2677
Set 74	**HSF-IF.C.7.d**	Graph rational functions with zeros and asymptotes	2714
Set 75	**HSF-IF.C.7.d**	Distinguish holes from vertical asymptotes	2751
Set 76	**HSF-BF.A.1.c**	Compose functions and interpret compositions with domains	2788
Set 77	**HSF-BF.B.4.b**	Verify inverses by composition on the stated domains	2825
Set 78	**HSF-BF.B.4.c**	Read values of an inverse function from graphs and tables	2862
Set 79	**HSF-BF.B.4.d**	Restrict a domain to make a function invertible and write the inverse	2899
Set 80	**HSF-BF.B.5**	Use the inverse relationship of exponents and logarithms to solve equations	2936
Set 81	**HSF-TF.A.3**	Derive values from special triangles and relate π - x, π + x, 2π - x to x	2973
Set 82	**HSF-TF.A.4**	Explain the symmetry and periodicity of trigonometric functions with the unit circ	3010
Set 83	**HSF-TF.B.6**	Choose and justify domain restrictions that make trigonometric functions invertib	3047
Set 84	**HSF-TF.B.7**	Solve trigonometric equations in models with inverse functions	3084

Write complex numbers in the form a + bi

MAIN	Algebra II main question	**HSN-CN.A.1**
BACKWARD 1	Simplify square roots of positive numbers	**HSN-RN.A.2**
BACKWARD 2	No real number squares to a negative number	**8.EE.A.2**
BACKWARD 3	Divide each term of a sum by a number	**6.EE.A.3**
FORWARD 1	Use equality of complex numbers to find unknown real values	**HSN-CN.A.1**
FORWARD 2	Represent complex numbers as points in the complex plane	**HSN-CN.B.4**

SET 1 • HSN-CN.A.1

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-49)}$ in the form bi, where b is a real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

7i

GRADING NOTE Form required: bi. $\sqrt{(-49)} = \sqrt{49} \cdot \sqrt{(-1)} = 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

Write √(−18) in the form bi with b in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

3i√2 (that is, 3√2 i)

GRADING NOTE Form required: simplest radical form. √(−18) = √9 · √2 · i = 3√2 i. Not accepted: √18 i (not simplified).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

Write $6 - \sqrt{(-20)}$ in the form $a + bi$. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

$6 - 2\sqrt{5}\,i$; real part 6, imaginary part $-2\sqrt{5}$

GRADING NOTE Form required: $a + bi$ with both parts named. $\sqrt{(-20)} = 2\sqrt{5}\,i$. The imaginary part is the real number $-2\sqrt{5}$, not $-2\sqrt{5}\,i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

Write (−4 + √(−75)) / 2 in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

Short answer

CORRECT ANSWER

−2 + (5√3/2)i; real part −2, imaginary part 5√3/2

GRADING NOTE Form required: a + bi with both parts named. √(−75) = 5√3 i; divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

Write (10 + √(−64)) / 4 in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

5/2 + 2i; real part 5/2, imaginary part 2

GRADING NOTE Form required: a + bi with both parts named (5/2 or 2.5 accepted). √(−64) = 8i; 10/4 = 5/2 and 8/4 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 1

Simplify square roots of positive numbers • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify √50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5√2

GRADING NOTE Form required: simplest radical form. 50 = 25 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q1**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

Simplify √72.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6√2

GRADING NOTE Form required: simplest radical form. 72 = 36 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q2**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

Which is √12 in simplest radical form?

A. 4√3

B. 3√2

C. 6√2

D. 2√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2√3

GRADING NOTE $12 = 4 \cdot 3$, so $\sqrt{12} = 2\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q3**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 7$. A square root of a sum is not the sum of the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

Simplify √28.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2√7

GRADING NOTE Form required: simplest radical form. 28 = 4 · 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q5**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 2

No real number squares to a negative number • Grade 8 • Expressions & Equations • 8.EE.A.2

Find all real numbers x with $x^2 = 81$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 9 or x = −9

GRADING NOTE Any order. Both 9^2 and $(-9)^2$ equal 81.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

True or false?

$(-3)^2 = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-3)^2 = (-3)(-3) = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

Is there a real number x with $x^2 = -4$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

Short answer

CORRECT ANSWER

No: the square of any real number is 0 or positive.

GRADING NOTE Any equivalent reason (a positive times a positive and a negative times a negative are both positive).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

Which number has a square equal to 36?

A. 18

B. $\sqrt{-36}$

C. -6

D. -18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −6

GRADING NOTE $(-6)^2 = 36$. $\sqrt{-36}$ is not a real number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(-7)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

Short answer

CORRECT ANSWER

49

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 3

Divide each term of a sum by a number • Grade 6 • Expressions & Equations • 6.EE.A.3

Write (8 + 6√2) / 2 as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

27

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 3√2

GRADING NOTE Any equivalent form. Divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (−6 + 4√5) / 2 as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

28

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−3 + 2√5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

True or false?

$(10 + \sqrt{7}) / 5 = 2 + \sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

29

QUESTION ID

S1-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both terms must be divided: $(10 + \sqrt{7})/5 = 2 + \sqrt{7}/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

(12 − 9√3) / 3 equals

A. 4 − 3√3

B. 4 − 9√3

C. 4 − 3√1

D. 12 − 3√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

30

QUESTION ID

S1-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4 − 3√3

GRADING NOTE 12/3 = 4 and 9√3/3 = 3√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

Write (−4 + 2√3) / (−2) as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

31

QUESTION ID

S1-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2 − √3

GRADING NOTE Any equivalent form. −4/(−2) = 2; 2√3/(−2) = −√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 1

Use equality of complex numbers to find unknown real values • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

x and y are real numbers and (x − 3) + 2yi = 5 − 8i.
Find x and y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

33

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 8, y = −4

GRADING NOTE Form required: both values. Equal complex numbers have equal real parts (x − 3 = 5) and equal imaginary parts (2y = −8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

a and b are real numbers and 3a + (b + 1)i = −12 + 4i. Find a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

34

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = −4, b = 3

GRADING NOTE Form required: both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

True or false?

If x and y are real and x + yi = 7, then y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

35

QUESTION ID

S1-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x + yi = 7 + 0i, so x = 7 and y = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

p and q are real and (p + q) + (p − q)i = 10 + 2i. What is p?

A. 6

B. 5

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

36

QUESTION ID

S1-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE p + q = 10 and p − q = 2 give p = 6 and q = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

m is a real number. For what value of m is $(m^2 - 4) + (m + 3)i$ a real number? What real number is it then?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

37

QUESTION ID

S1-F1-Q5

TYPE

Short answer

CORRECT ANSWER

m = −3; the number is 5

GRADING NOTE Form required: both. The imaginary part m + 3 must be 0, so m = −3 and $m^2 - 4 = 9 - 4 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 2

Represent complex numbers as points in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

In the complex plane the horizontal axis is real and the vertical axis is imaginary. Give the point for −3 + 5i.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

39

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, 5)

GRADING NOTE Form required: an ordered pair (real part, imaginary part).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

Which complex number is represented by the point (4, −2) in the complex plane?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

40

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 − 2i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

Find the distance from 0 to the point 6 + 8i in the complex plane.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

41

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $\sqrt{(6^2 + 8^2)} = 10$ (the modulus $|6 + 8i|$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number lies on the horizontal axis of the complex plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

42

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real number a is a + 0i, the point (a, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

Which number lies on the vertical axis of the complex plane?

A. -7

B. $7 + i$

C. $-7i$

D. $1 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

43

QUESTION ID
S1-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. $-7i$

GRADING NOTE $-7i = 0 - 7i$, the point $(0, -7)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

Add and subtract complex numbers

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Add and subtract integers and decimals	**7.NS.A.1**
BACKWARD 3	Write square roots of negative numbers with i	**HSN-CN.A.1**
FORWARD 1	Find real parameters that make a sum real or imaginary	**HSN-CN.A.2**
FORWARD 2	Add complex numbers as vectors in the complex plane	**HSN-CN.B.5**

SET 2 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Simplify (3 + 5i) + (−7 + 2i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

46

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

−4 + 7i

GRADING NOTE Form required: a + bi. 3 − 7 = −4 and 5i + 2i = 7i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

Simplify (6 − 4i) − (−2 + 9i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

47

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

8 − 13i

GRADING NOTE Form required: a + bi. Subtract both parts: 6 − (−2) = 8 and −4i − 9i = −13i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

Simplify (−1 + √(−16)) + (5 − √(−9)). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

48

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

4 + i

GRADING NOTE Form required: a + bi. (−1 + 4i) + (5 − 3i) = 4 + i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

$z_1 = 2.5 - i$, $z_2 = -1.5 + 3i$ and $z_3 = 4i$. Find $z_1 - z_2 + z_3$ in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

49

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

4 (that is, 4 + 0i)

GRADING NOTE Form required: a + bi or the real number 4. Real: 2.5 + 1.5 = 4; imaginary: −1 − 3 + 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

Find the complex number w for which
(4 − 7i) + w = −2 + 3i.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

50

QUESTION ID
S2-M-Q5

TYPE
Short answer

CORRECT ANSWER
−6 + 10i

GRADING NOTE Form required: a + bi. w = (−2 + 3i) − (4 − 7i) = −6 + 10i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (3 + 5x) + (−7 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

52

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−4 + 7x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

Simplify (6 − 4x) − (−2 + 9x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

53

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8 − 13x

GRADING NOTE Any equivalent form. The minus sign applies to both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

(2 − 3y) − (5 − y) equals

A. −3 − 2y

B. 7 − 4y

C. −3 + 2y

D. −3 − 4y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

54

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −3 − 2y

GRADING NOTE 2 − 5 = −3 and −3y + y = −2y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$(4 + 2t) + (1 - 6t) = 5 + 4t.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

55

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2t - 6t = -4t$, so the sum is $5 - 4t$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (0.5 − x) − (1.5 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

56

QUESTION ID

S2-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−1 − 3x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 2

Add and subtract integers and decimals • Grade 7 • The Number System • 7.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $6 - (-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

58

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-4 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

59

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$-1 - 3 + 4$ equals

A. 8

B. 0

C. -8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

60

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0

GRADING NOTE $-1 - 3 = -4$ and $-4 + 4 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 - (-1.5) = 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

61

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2.5 + 1.5 = 4.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

62

QUESTION ID

S2-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 3

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−16) in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

64

QUESTION ID

S2-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $\sqrt{(-9)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

65

QUESTION ID

S2-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

√(−25) equals

A. −5

B. 25i

C. −5i

D. 5i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

66

QUESTION ID

S2-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5i

GRADING NOTE √(−25) = √25 · i = 5i (the principal root).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{(-1)} = i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

67

QUESTION ID

S2-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of i as the principal square root of −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $-\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

68

QUESTION ID

S2-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−6i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 1

Find real parameters that make a sum real or imaginary • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

k is a real number. For what value of k is (3 + ki) + (k − 5i) a real number? Give that real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

70

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

k = 5; the sum is 8

GRADING NOTE Form required: k and the number. The imaginary part k − 5 must be 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

k is real. For what value of k is
(2k − 6) + 4i − (k + 1 − i) a pure imaginary number
(real part 0)?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

71

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. Real part: 2k − 6 − k − 1 = k − 7 = 0. The sum is then 5i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

True or false?

The sum of a complex number and its conjugate is always 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

72

QUESTION ID

S2-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (a + bi) + (a − bi) = 2a, which is 0 only when a = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

a is real and (a + 3i) − (2 + ai) is real. What is a?

A. 0

B. 2

C. 3

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

73

QUESTION ID

S2-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE Imaginary part 3 − a = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

Explain why the sum of a + bi and a − bi is always a real number when a and b are real.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

74

QUESTION ID

S2-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The imaginary parts b and −b add to 0, leaving 2a, a real number.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q5**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 2

Add complex numbers as vectors in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5

Plot 2 + i and 1 + 3i as points in the complex plane (real axis horizontal). What point represents their sum?

Sum point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

76

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 4), the number 3 + 4i

GRADING NOTE Form required: the point or the number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

The points 0, 2 + i, 1 + 3i and their sum are vertices of a quadrilateral. What kind of quadrilateral is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

77

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A parallelogram

GRADING NOTE Any equivalent answer. Adding complex numbers adds their vectors (parallelogram rule).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

True or false?

Subtracting w from z gives the vector from the point w to the point z.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

78

QUESTION ID
S2-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE z − w is the displacement from w to z.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

What is the distance between the points 5 + 2i and 1 − i in the complex plane?

A. 25

B. √13

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

79

QUESTION ID

S2-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 5

GRADING NOTE |(5 + 2i) − (1 − i)| = |4 + 3i| = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

Find the midpoint of the segment joining $6 - 2i$ and $-2 + 4i$, as a complex number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

80

QUESTION ID

S2-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2 + i$

GRADING NOTE Form required: a + bi. The average $((6 - 2i) + (-2 + 4i))/2 = 2 + i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

Multiply complex numbers and simplify powers of i

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Multiply binomials	**HSA-APR.A.1**
BACKWARD 2	Use $i^2 = -1$ and the cycle of powers of i	**HSN-CN.A.1**
BACKWARD 3	Divide whole numbers to find a remainder	**4.NBT.B.6**
FORWARD 1	Find a real parameter that makes a product real	**HSN-CN.A.2**
FORWARD 2	Divide complex numbers using conjugates	**HSN-CN.A.3**

SET 3 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply (3 + 2i)(4 − 5i). Write the product in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

83

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

22 − 7i

GRADING NOTE Form required: a + bi. $12 - 15i + 8i - 10i^2 = 12 - 7i + 10 = 22 - 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

Simplify $(-2 + i)^2$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

84

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

3 − 4i

GRADING NOTE Form required: a + bi. $4 - 4i + i^2 = 4 - 4i - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

Multiply $(5 - 3i)(5 + 3i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

85

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

34 (that is, 34 + 0i)

GRADING NOTE Form required: $a + bi$ or the real number 34. $25 + 15i - 15i - 9i^2 = 25 + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

Simplify $(1 + i)^3$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

86

QUESTION ID

S3-M-Q4

TYPE

Short answer

CORRECT ANSWER

−2 + 2i

GRADING NOTE Form required: a + bi. $(1 + i)^2 = 2i$, and $2i(1 + i) = 2i + 2i^2 = -2 + 2i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

Simplify $i^{23} \cdot (4 - i)$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

87

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

−1 − 4i

GRADING NOTE Form required: a + bi. 23 = 4 · 5 + 3, so $i^{23} = i^3 = -i$; $-i(4 - i) = -4i + i^2 = -1 - 4i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 1

Multiply binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply (3 + 2x)(4 − 5x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

89

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$12 - 7x - 10x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(-2 + x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

90

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$4 - 4x + x^2$

GRADING NOTE Any equivalent form. Not $x^2 + 4$ (missing the middle term).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

(5 − 3x)(5 + 3x) equals

A. $25 + 9x^2$

B. $10 − 9x^2$

C. $25 − 9x^2$

D. $25 − 30x − 9x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

91

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $25 − 9x^2$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$(1 + x)^2 = 1 + x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

92

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(1 + x)^2 = 1 + 2x + x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

Multiply $x(4 - x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

93

QUESTION ID

S3-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$4x - x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 2

Use $i^2 = -1$ and the cycle of powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

95

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. Definition of i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

Simplify i^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

96

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-i$

GRADING NOTE Any equivalent form. $i^3 = i^2 \cdot i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

Simplify i^4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

97

QUESTION ID

S3-B2-Q3

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. $i^4 = (i^2)^2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-10i^2 = -10.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

98

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^2 = -1$, so $-10i^2 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

Which equals 1?

A. i^8

B. i^9

C. i^3

D. i^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

99

QUESTION ID

S3-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. i^8

GRADING NOTE Powers of i repeat every 4; 8 is a multiple of 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 3

Divide whole numbers to find a remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find the remainder when 23 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

101

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 23 = 4 · 5 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

Find the remainder when 50 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

102

QUESTION ID

S3-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 50 = 4 · 12 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

What is the remainder when 36 is divided by 4?

A. 0

B. 4

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

103

QUESTION ID

S3-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0

GRADING NOTE $36 = 4 \cdot 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

True or false?

The remainder when 101 is divided by 4 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

104

QUESTION ID

S3-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $101 = 4 \cdot 25 + 1$, so the remainder is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

Write 39 as 4q + r with 0 ≤ r < 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

105

QUESTION ID

S3-B3-Q5

TYPE

Short answer

CORRECT ANSWER

39 = 4 · 9 + 3

GRADING NOTE Form required: q = 9 and r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 1

Find a real parameter that makes a product real • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

a is a real number. For what value of a is (a + 2i)(3 − i) a real number? Give the product.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

107

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

a = 6; the product is 20

GRADING NOTE Form required: both. (a + 2i)(3 − i) = (3a + 2) + (6 − a)i; the imaginary part 6 − a = 0. Product 18 + 2 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

b is real. For what value of b is (2 + bi)(1 + i) pure imaginary (real part 0)?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

108

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Product: (2 − b) + (2 + b)i; real part 2 − b = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

True or false?

The product of two non-real complex numbers can be a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

109

QUESTION ID

S3-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example (1 + i)(1 − i) = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

For real c, $(c + i)^2$ is real when

A. $c = 0$

B. $c = 1$

C. $c = -1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

110

QUESTION ID

S3-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. c = 0

GRADING NOTE $(c + i)^2 = c^2 - 1 + 2ci$; imaginary part $2c = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q4**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

Show that (a + bi)(a − bi) is a real number for all real a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

111

QUESTION ID

S3-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$a^2 - abi + abi - b^2i^2 = a^2 + b^2$, which is real.

GRADING NOTE Any equivalent algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 2

Divide complex numbers using conjugates • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Write (3 + 4i) / (1 + 2i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

113

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

11/5 − (2/5)i

GRADING NOTE Form required: a + bi. Multiply by (1 − 2i)/(1 − 2i): (3 + 4i)(1 − 2i)/5 = (11 − 2i)/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

Write 10 / (3 − i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

114

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 + i

GRADING NOTE Form required: a + bi. 10(3 + i)/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

True or false?

To divide by 2 − 5i, multiply numerator and denominator by 2 − 5i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

115

QUESTION ID

S3-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the conjugate 2 + 5i; $(2 - 5i)^2$ is not real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

(2 + i) / i equals

A. −1 + 2i

B. 2 + i

C. 1 − 2i

D. 1 + 2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

116

QUESTION ID

S3-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 − 2i

GRADING NOTE (2 + i)(−i)/(i · −i) = (−2i + 1)/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

Explain why multiplying by the conjugate of the denominator always gives a real denominator.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

117

QUESTION ID

S3-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$(c + di)(c - di) = c^2 + d^2$, a real number (and positive unless $c = d = 0$).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSN-CN.C.7 • THE COMPLEX NUMBER SYSTEM

Solve quadratic equations with complex solutions

MAIN	Algebra II main question	**HSN-CN.C.7**
BACKWARD 1	Use the quadratic formula and the discriminant	**HSA-REI.B.4.b**
BACKWARD 2	Write square roots of negative numbers with i	**HSN-CN.A.1**
BACKWARD 3	Solve equations of the form $(x - h)^2 = k$	**HSA-REI.B.4.b**
FORWARD 1	Find a quadratic from one complex solution	**HSN-CN.C.7**
FORWARD 2	Classify solutions for a family of quadratics	**HSN-CN.C.7**

SET 4 • HSN-CN.C.7

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 49 = 0$ over the complex numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

120

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7i or x = −7i

GRADING NOTE Form required: both solutions, exact. $x^2 = -49$, $x = \pm 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-M-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

Solve $x^2 - 4x + 13 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

121

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2 + 3i or

x = 2 − 3i

GRADING NOTE Form required: both, a + bi. Discriminant 16 − 52 = −36; x = (4 ± 6i)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

Solve $2x^2 + 2x + 5 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

122

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = −1/2 + (3/2)i or
x = −1/2 − (3/2)i

GRADING NOTE Form required: both, a + bi. Discriminant 4 − 40 = −36; x = (−2 ± 6i)/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

Solve $(x + 3)^2 = -20$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

123

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

$x = -3 + 2\sqrt{5}\,i$ or
$x = -3 - 2\sqrt{5}\,i$

GRADING NOTE Form required: both, exact. $x + 3 = \pm\sqrt{-20} = \pm 2\sqrt{5}\,i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

Solve $3x^2 - 6x + 7 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

124

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 1 + (2\sqrt{3}/3)i$ or
$x = 1 - (2\sqrt{3}/3)i$

GRADING NOTE Form required: both, exact, a + bi. Discriminant $36 - 84 = -48 = (4\sqrt{3}\,i)^2$; $x = (6 \pm 4\sqrt{3}\,i)/6$. Check: product 7/3 and sum 2 agree with c/a and −b/a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 1

Use the quadratic formula and the discriminant • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Find the discriminant $b^2 - 4ac$ of $x^2 - 4x + 13 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

126

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−36

GRADING NOTE Any equivalent form. 16 − 52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

Find the discriminant of $2x^2 + 2x + 5 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

127

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−36

GRADING NOTE Any equivalent form. 4 − 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

True or false?

If the discriminant is negative, the quadratic equation has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

128

QUESTION ID

S4-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A negative discriminant gives two non-real solutions and no real ones.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In $3x^2 - 6x + 7 = 0$, the values of a, b and c are

A. 3, −6, −7

B. 3, 6, 7

C. −3, −6, 7

D. 3, −6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

129

QUESTION ID

S4-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3, −6, 7

GRADING NOTE Standard form $ax^2 + bx + c = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

Write the quadratic formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

130

QUESTION ID

S4-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 2

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−36) in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

132

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−48) in simplest form with i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

133

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4√3 i

GRADING NOTE Form required: simplest radical form. 48 = 16 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

√(−20) equals

A. 2√5 i

B. 4√5 i

C. −2√5

D. 10i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

134

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2√5 i

GRADING NOTE √20 = 2√5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{(-49)} = -7.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

135

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(-49)} = 7i.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

Write (−2 + √(−36)) / 4 in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

136

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2 + (3/2)i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 3

Solve equations of the form $(x - h)^2 = k$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 2)^2 = 9$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

138

QUESTION ID

S4-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = 5 or x = −1

GRADING NOTE Form required: both. $x - 2 = \pm 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

Solve $(x + 1)^2 = 5$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

139

QUESTION ID

S4-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = −1 + √5 or
x = −1 − √5

GRADING NOTE Form required: both, exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

True or false?

Solving $x^2 = 16$ gives only $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

140

QUESTION ID

S4-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Also $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

The solutions of $(x + 4)^2 = 25$ are

A. 5 and −5

B. −1 and 9

C. 1 and −9

D. 21 and −29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

141

QUESTION ID

S4-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 and −9

GRADING NOTE $x + 4 = \pm 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

Solve $x^2 = 50$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

142

QUESTION ID

S4-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 5\sqrt{2}$ or $x = -5\sqrt{2}$

GRADING NOTE Form required: both, exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B3-Q5**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 1

Find a quadratic from one complex solution • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

$3 - 2i$ is a solution of $x^2 + bx + c = 0$, where b and c are real. Find b and c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

144

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

b = −6, c = 13

GRADING NOTE Form required: both. With real coefficients the other solution is 3 + 2i; sum 6 = −b, product 9 + 4 = 13 = c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

$-1 + 4i$ is a solution of a quadratic equation $x^2 + bx + c = 0$ with real b and c. Write the equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

145

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x + 17 = 0$

GRADING NOTE Any equivalent equation. Sum −2, product 1 + 16 = 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

True or false?

If 2 + i is a solution of a quadratic with real coefficients, then 2 − i is also a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

146

QUESTION ID

S4-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Non-real solutions of real quadratics come in conjugate pairs (± in the formula).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

Which quadratic with real coefficients has 5i as a solution?

A. $x^2 + 25 = 0$

B. $x^2 + 5 = 0$

C. $x^2 - 25 = 0$

D. $x^2 - 10x + 25 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

147

QUESTION ID

S4-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 25 = 0$

GRADING NOTE $(5i)^2 + 25 = -25 + 25 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

Explain why $x^2 - (2 + i)x = 0$ has the solution $2 + i$ but not the solution $2 - i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

148

QUESTION ID

S4-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Its coefficient 2 + i is not real, so solutions need not come in conjugate pairs: the solutions are 0 and 2 + i.

GRADING NOTE Any equivalent explanation naming the non-real coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 2

Classify solutions for a family of quadratics • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

For which real values of k does $x^2 + kx + 9 = 0$ have two non-real solutions?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

150

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-6 < k < 6$

GRADING NOTE Form required: an inequality. Discriminant $k^2 - 36 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

For which real values of c does $x^2 - 4x + c = 0$ have non-real solutions? Give them for c = 8.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

151

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

c > 4; for c = 8, x = 2 ± 2i

GRADING NOTE Form required: both parts. Discriminant $16 - 4c < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

True or false?

$x^2 + kx - 1 = 0$ has non-real solutions for some real k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

152

QUESTION ID

S4-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its discriminant $k^2 + 4$ is always positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

$x^2 + 6x + m = 0$ has exactly one (repeated) real solution when m =

A. 3

B. 36

C. −9

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

153

QUESTION ID

S4-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9

GRADING NOTE $36 - 4m = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q4**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

For c > 4, write the solutions of $x^2 - 4x + c = 0$ in terms of c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

154

QUESTION ID

S4-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 2 \pm i\sqrt{(c - 4)}$

GRADING NOTE Form required: exact in terms of c. $(4 \pm \sqrt{(16 - 4c)})/2 = 2 \pm \sqrt{(4 - c)} = 2 \pm i\sqrt{(c - 4)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

Factor polynomials over the complex numbers

MAIN	Algebra II main question	**HSN-CN.C.8**
BACKWARD 1	Factor a difference of squares	**HSA-SSE.A.2**
BACKWARD 2	Multiply complex conjugates	**HSN-CN.A.2**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Factor cubic and quartic polynomials over the complex numbers	**HSN-CN.C.8**
FORWARD 2	Write polynomials with real coefficients from complex zeros	**HSN-CN.C.9**

SET 5 • HSN-CN.C.8

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

157

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 5i)(x - 5i)$

GRADING NOTE Any order of factors. $(x + 5i)(x - 5i) = x^2 - 25i^2 = x^2 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

Factor $x^2 + 7$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

158

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + i\sqrt{7})(x - i\sqrt{7})$

GRADING NOTE Any order. $x^2 - (i\sqrt{7})^2 = x^2 + 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

Factor $4x^2 + 9$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

159

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(2x + 3i)(2x - 3i)$

GRADING NOTE Any equivalent factorization (for example $4(x + 3i/2)(x - 3i/2)$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

Factor $x^4 - 16$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

160

QUESTION ID

S5-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 2)(x - 2i)(x + 2i)$

GRADING NOTE Any order. $x^4 - 16 = (x^2 - 4)(x^2 + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

Factor $x^2 - 6x + 13$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

161

QUESTION ID

S5-M-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - (3 + 2i))(x - (3 - 2i))$

GRADING NOTE Any equivalent form. Its zeros are $3 \pm 2i$ (discriminant -16). Check: sum 6, product $9 + 4 = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 1

Factor a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 49$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

163

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 7)(x − 7)

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

Factor $9x^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

164

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(3x + 2)(3x − 2)

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

$x^4 - 1$ equals

A. $(x - 1)^4$

B. $(x^2 - 1)^2$

C. $(x^2 - 1)(x^2 + 1)$

D. $(x^2 + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

165

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x^2 - 1)(x^2 + 1)$

GRADING NOTE $(x^2)^2 - 1^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

166

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

Factor $x^2 - 5$ using radicals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

167

QUESTION ID

S5-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + \sqrt{5})(x - \sqrt{5})$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 2

Multiply complex conjugates • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(x + 5i)(x - 5i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

169

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 25$

GRADING NOTE Any equivalent form. $-25i^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(2x + 3i)(2x - 3i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

170

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$4x^2 + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

True or false?

$(3i)^2 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

171

QUESTION ID

S5-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(3i)^2 = 9i^2 = -9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

$(x - (1 + i))(x - (1 - i))$ equals

A. $x^2 + 2x + 2$

B. $x^2 - 2x + 2$

C. $x^2 + 2$

D. $x^2 - 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

172

QUESTION ID

S5-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x^2 - 2x + 2$

GRADING NOTE Sum 2, product 1 + 1 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

Simplify $(i\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

173

QUESTION ID

S5-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q5**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 25 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

175

QUESTION ID

S5-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = ±5i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B3-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

Solve $x^2 - 6x + 13 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

176

QUESTION ID

S5-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 3 \pm 2i$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B3-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

True or false?

If r is a zero of p(x), then x − r is a factor of p(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

177

QUESTION ID

S5-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Factor theorem, also for complex r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

The zeros of $x^2 + 4$ are

A. 2i and −2i

B. 4i and −4i

C. −4

D. 2 and −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

178

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2i and −2i

GRADING NOTE $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x + 5 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

179

QUESTION ID

S5-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x = −1 ± 2i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 1

Factor cubic and quartic polynomials over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^3 + 8$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

181

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 2)(x - 1 - i\sqrt{3})(x - 1 + i\sqrt{3})$

GRADING NOTE Any order. $x^3 + 8 = (x + 2)(x^2 - 2x + 4)$, and $x^2 - 2x + 4$ has zeros $1 \pm i\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^3 + 9x$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

182

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

Factor $x^4 + 5x^2 + 4$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

183

QUESTION ID

S5-F1-Q3

TYPE

Short answer

CORRECT ANSWER

(x + i)(x − i)(x + 2i)(x − 2i)

GRADING NOTE Any order. $(x^2 + 1)(x^2 + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

True or false?

$x^3 - 1 = (x - 1)(x^2 + x + 1)$, and $x^2 + x + 1$ has two non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

184

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Discriminant $1 - 4 = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

Which is a linear factor of $x^3 - 27$ over the complex numbers?

A. $x + 3i$

B. $x + 3$

C. $x - 3i$

D. $x - (-3/2 + (3\sqrt{3}/2)i)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

185

QUESTION ID

S5-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D.

$x - (-3/2 + (3\sqrt{3}/2)i)$

GRADING NOTE $x^3 - 27 = (x - 3)(x^2 + 3x + 9)$; $x^2 + 3x + 9$ has zeros $-3/2 \pm (3\sqrt{3}/2)i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 2

Write polynomials with real coefficients from complex zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Write a polynomial with real coefficients, leading coefficient 1 and least degree, whose zeros include 2i and 1 − i.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

187

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 4)(x^2 - 2x + 2) = x^4 - 2x^3 + 6x^2 - 8x + 8$

GRADING NOTE Any equivalent form. Conjugates −2i and 1 + i must also be zeros, so degree 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

What is the least degree of a polynomial with real coefficients that has zeros 3, 4i and 2 + i?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

188

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. −4i and 2 − i are also zeros: 1 + 2 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

True or false?

A polynomial with real coefficients can have exactly one non-real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

189

QUESTION ID

S5-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Non-real zeros come in conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

A cubic with real coefficients has zeros 5 and −i. Its third zero is

A. −5

B. 5i

C. 1

D. i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

190

QUESTION ID

S5-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. i

GRADING NOTE Conjugate of −i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

Explain why a polynomial with real coefficients and degree 3 must have at least one real zero.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

191

QUESTION ID

S5-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It has 3 complex zeros with multiplicity; non-real zeros pair up, so they come two at a time, leaving at least one real zero.

GRADING NOTE Any equivalent explanation using the Fundamental Theorem of Algebra and conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

SET 6 • HSN-CN.C.9 • THE COMPLEX NUMBER SYSTEM

Use the Fundamental Theorem of Algebra

MAIN	Algebra II main question	**HSN-CN.C.9**
BACKWARD 1	Find the degree of a polynomial	**HSA-SSE.A.1.a**
BACKWARD 2	Find zeros from factored form	**HSA-SSE.B.3.a**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Relate the complex zeros to the real x-intercepts of a graph	**HSA-APR.B.3**
FORWARD 2	Prove existence statements about real zeros	**HSN-CN.C.9**

SET 6 • HSN-CN.C.9

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does $p(x) = 3x^5 - x^2 + 7$ have? Name the theorem you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

194

QUESTION ID
S6-M-Q1

TYPE
Short answer

CORRECT ANSWER
5, by the Fundamental Theorem of Algebra (degree 5)

GRADING NOTE Form required: the number and the theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

List every complex zero of $p(x) = (x - 2)^3(x^2 + 9)$ with its multiplicity, and check that the multiplicities add to the degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

195

QUESTION ID
S6-M-Q2

TYPE
Short answer

CORRECT ANSWER
2 (multiplicity 3), 3i (multiplicity 1), −3i (multiplicity 1);
3 + 1 + 1 = 5
= degree

GRADING NOTE Form required: all zeros with multiplicities and the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

Show that the Fundamental Theorem of Algebra holds for every quadratic $p(x) = ax^2 + bx + c$ with real a, b, c and $a \neq 0$: p has exactly two complex zeros counted with multiplicity.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

196

QUESTION ID

S6-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The zeros are $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$: two real zeros if $D > 0$, one real zero of multiplicity 2 if $D = 0$, two conjugate non-real zeros if $D < 0$.

GRADING NOTE REQUIRED WORK: Use the quadratic formula and the cases of the discriminant.

Any valid proof or derivation. Essential: the formula gives all zeros; the three cases of $D = b^2 - 4ac$; for $D < 0$, $\sqrt{D} = i\sqrt{(-D)}$ is a complex number; for $D = 0$ the zero $-b/(2a)$ has multiplicity 2 since $p(x) = a(x + b/(2a))^2$. Invalid: checking one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

Find every complex zero of $p(x) = x^4 - 81$ and check that the count agrees with the Fundamental Theorem of Algebra.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

197

QUESTION ID

S6-M-Q4

TYPE

Short answer

CORRECT ANSWER

3, −3, 3i, −3i; four zeros for degree 4

GRADING NOTE Form required: all four and the check. $(x^2 - 9)(x^2 + 9)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

A student says $p(x) = x^3 + x + 1$ has no zeros because it does not factor with integers. Use the Fundamental Theorem of Algebra to explain how many complex zeros it has, and explain why at least one of them is real.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

198

QUESTION ID

S6-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

It has 3 complex zeros with multiplicity; non-real zeros of a real polynomial come in conjugate pairs, so at most 2 are non-real and at least 1 is real.

GRADING NOTE REQUIRED WORK: Count zeros by degree; use conjugate pairs for real coefficients.

Any valid explanation. Essential: degree 3 gives 3 zeros; "no integer factorization" does not mean "no zeros"; conjugate pairs force an odd number of real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 1

Find the degree of a polynomial • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

What is the degree of $3x^5 - x^2 + 7$?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

200

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

What is the degree of $(x - 2)^3(x^2 + 9)$ when multiplied out?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

201

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 3 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

The degree of $x(x + 1)(x^2 - 4)$ is

A. 1

B. 4

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

202

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE 1 + 1 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

True or false?

The degree of $4x^3 + 2x^4$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

203

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The highest power is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

What is the leading coefficient of $-2x^3 + 5x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

204

QUESTION ID

S6-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 2

Find zeros from factored form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the zeros of $(x - 2)(x + 5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

206

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 and −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B2-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

Find the zeros of $(x - 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

207

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1 (repeated three times)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

True or false?

A product is 0 only when one of its factors is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

208

QUESTION ID

S6-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Zero-product property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

The zeros of x(x + 4) are

A. 0 and 4

B. 0 and −4

C. −4 only

D. 4 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

209

QUESTION ID

S6-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0 and −4

GRADING NOTE x = 0 or x + 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the zeros of $(2x - 3)(x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

210

QUESTION ID

S6-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3/2 and −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 9 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

212

QUESTION ID

S6-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = ±3i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B3-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x + 2 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

213

QUESTION ID

S6-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = −1 ± i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

True or false?

$x^2 + 1 = 0$ has no complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

214

QUESTION ID

S6-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has ±i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

The solutions of $x^2 - 2x + 5 = 0$ are

A. $1 \pm 4i$

B. $-1 \pm 2i$

C. $1 \pm 2i$

D. $2 \pm i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

215

QUESTION ID

S6-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 ± 2i

GRADING NOTE Discriminant −16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = -81$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

216

QUESTION ID

S6-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x = ±9i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 1

Relate the complex zeros to the real x-intercepts of a graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A polynomial of degree 4 with real coefficients has a graph that crosses the x-axis at exactly two points and touches it nowhere else. Each crossing is a simple zero. How many non-real zeros does it have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

218

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 4 zeros in all, 2 real (the crossings), so 2 non-real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

The graph of $y = x^3 - 4x^2 + 5x - 2$ touches the x-axis at $x = 1$ and crosses it at $x = 2$. List all complex zeros with multiplicity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

219

QUESTION ID
S6-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1 (multiplicity 2) and 2 (multiplicity 1)

GRADING NOTE Form required: zeros with multiplicities. $(x - 1)^2(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 6 whose graph never meets the x-axis has 6 non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

220

QUESTION ID

S6-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All 6 zeros are non-real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

A degree-5 polynomial with real coefficients has exactly one real zero, of multiplicity 1. How many non-real zeros does it have?

A. 4

B. 5

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

221

QUESTION ID

S6-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE $5 - 1 = 4$ (two conjugate pairs).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

Can a degree-4 polynomial with real coefficients have exactly three real zeros, each of multiplicity 1? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

222

QUESTION ID

S6-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No: the fourth zero would be a single non-real zero, but non-real zeros come in conjugate pairs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 2

Prove existence statements about real zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Prove that every polynomial of odd degree with real coefficients has at least one real zero.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

224

QUESTION ID

S6-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

An odd number of zeros, minus an even number of non-real zeros, leaves an odd (so nonzero) number of real zeros.

GRADING NOTE

REQUIRED WORK: Count zeros with the theorem; pair non-real zeros.

Any valid proof or derivation. Essential: n zeros with multiplicity (n odd); conjugate pairs have equal multiplicity, so the non-real zeros count is even; odd minus even is odd ≥ 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

A degree-4 polynomial with real coefficients has the zeros 1 + i and 3. What are all of its possible zero lists?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

225

QUESTION ID
S6-F2-Q2

TYPE
Short answer

CORRECT ANSWER
1 + i, 1 − i, 3 and one more real zero r (r may be 3 again, giving 3 with multiplicity 2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

True or false?

A degree-4 polynomial with real coefficients must have at least one real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

226

QUESTION ID
S6-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $x^4 + 1$ has no real zeros; only odd degree guarantees a real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

A polynomial with real coefficients has degree 7. The possible numbers of real zeros (with multiplicity) are

A. 1, 3, 5, 7

B. 7 only

C. 0, 2, 4, 6

D. 0 to 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

227

QUESTION ID

S6-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1, 3, 5, 7

GRADING NOTE The number of real zeros is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

Explain why the Fundamental Theorem of Algebra says nothing about whether a polynomial's zeros can be found exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

228

QUESTION ID
S6-F2-Q5

TYPE
Short answer

CORRECT ANSWER
It guarantees that the zeros exist and counts them; it gives no method or formula for finding them.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a • SEEING STRUCTURE IN EXPRESSIONS

Interpret the parts of polynomial, rational, and exponential expressions in context

MAIN	Algebra II main question	**HSA-SSE.A.1.a**
BACKWARD 1	Name terms, factors, and coefficients	**HSA-SSE.A.1.a**
BACKWARD 2	Translate between a percent change and a multiplying factor	**7.RP.A.3**
BACKWARD 3	Interpret a quotient as an amount per unit	**6.RP.A.2**
FORWARD 1	Build an expression whose parts have given meanings	**HSA-SSE.A.1.a**
FORWARD 2	Rewrite an exponential expression to reveal a different rate	**HSF-IF.C.8.b**

SET 7 • HSA-SSE.A.1.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A town's population is modeled by $P(t) = 12{,}500(0.97)^t$, where t is the number of years after 2020. What do 12,500 and 0.97 represent, and what does $1 - 0.97$ represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

231

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

12,500 is the population in 2020; 0.97 is the yearly factor (each year the population is 97% of the year before); 1 − 0.97 = 0.03 means the population decreases by 3% per year.

GRADING NOTE Any equivalent interpretation of all three parts, with units (people, per year).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The average cost per item of making x items is C(x) = (4,000 + 6x) / x dollars. Interpret 4,000, 6x, and the division by x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

232

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

4,000 is a fixed cost in dollars; 6x is the variable cost ($6 per item for x items); dividing the total cost by x gives the cost per item.

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

An open box is made by cutting squares of side x cm from the corners of a 20 cm by 20 cm sheet. Its volume is $V(x) = x(20 - 2x)^2$ cm³. Interpret x, $20 - 2x$, and $(20 - 2x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

233

QUESTION ID

S7-M-Q3

TYPE

Short answer

CORRECT ANSWER

x is the height of the box; $20 - 2x$ is the side length of the square base (20 cm minus two cuts); $(20 - 2x)^2$ is the base area in cm².

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The amount of a medicine in the body t hours after a dose is $A(t) = 200(1/2)^{t/6}$ mg. Interpret 200, 1/2, and t/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

234

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

200 mg is the initial dose; the amount is halved repeatedly; t/6 counts 6-hour periods, so the amount halves every 6 hours (half-life 6 hours).

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

Maya keeps $1,500 in an account that earns 4% interest compounded once a year and also puts $200 per year in a jar (no interest). Her total after n years is $T(n) = 1500(1.04)^n + 200n$. Interpret $1500(1.04)^n$, 1.04, and 200n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

235

QUESTION ID
S7-M-Q5

TYPE
Short answer

CORRECT ANSWER
$1500(1.04)^n$ is the account balance after n years; 1.04 is the yearly growth factor (4% interest added each year); 200n is the cash in the jar after n years.

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 1

Name terms, factors, and coefficients • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

Name the terms of 4,000 + 6x.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q1/5

237

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4,000 and 6x

GRADING NOTE Form required: both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q1/5

What is the coefficient of x in 7x − 3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q2/5

238

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q2/5

In 5x(x − 2), which is a factor?

A. 5x − 2

B. x − 10

C. x − 2

D. −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q3/5

239

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x − 2

GRADING NOTE 5x(x − 2) is the product of 5, x and x − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q3/5

True or false?

In 3 + 2x, the number 3 is a factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q4/5

240

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 is a term (it is added).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the factors of $200(0.5)^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q5/5

241

QUESTION ID

S7-B1-Q5

TYPE

Short answer

CORRECT ANSWER

200 and $(0.5)^t$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 2

Translate between a percent change and a multiplying factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A price is multiplied by 0.97. By what percent does it change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q1/5

243

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

It decreases by 3%.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What factor multiplies an amount that increases by 4%?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q2/5

244

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1.04

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q2/5

Multiplying by 1.25 is the same as

A. increasing by 2.5%

B. increasing by 25%

C. increasing by 125%

D. decreasing by 25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q3/5

245

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. increasing by 25%

GRADING NOTE 1 + 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q3/5

True or false?

Multiplying by 0.8 decreases an amount by 80%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q4/5

246

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 20%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q4/5

Find 1 − 0.85 and explain what it means for an amount multiplied by 0.85.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q5/5

247

QUESTION ID

S7-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.15; the amount decreases by 15%.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 3

Interpret a quotient as an amount per unit • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A total cost of $600 is shared by 30 people. What does 600/30 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q1/5

249

QUESTION ID

S7-B3-Q1

TYPE

Short answer

CORRECT ANSWER

The cost per person, $20.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q1/5

A car travels 150 miles in 3 hours. What does 150/3 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q2/5

250

QUESTION ID

S7-B3-Q2

TYPE

Short answer

CORRECT ANSWER

The average speed, 50 miles per hour.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q2/5

If T is a total cost for n items, then T/n is

A. the profit

B. the number of items

C. the cost per item

D. the total cost

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q3/5

251

QUESTION ID

S7-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. the cost per item

GRADING NOTE Unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Dividing a total by the number of items gives an average per item.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q4/5

252

QUESTION ID

S7-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of a mean or unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q4/5

What does (4,000 + 6x)/x mean when x = 100?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q5/5

253

QUESTION ID

S7-B3-Q5

TYPE

Short answer

CORRECT ANSWER

The average cost per item when 100 items are made: $46.

GRADING NOTE Any equivalent answer. 4,600/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 1

Build an expression whose parts have given meanings • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Write an expression for the value of a $24,000 car that loses 15% of its value each year, t years after purchase.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q1/5

255

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$24{,}000(0.85)^t$

GRADING NOTE Any equivalent expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges a $90 joining fee and $35 per month. Write an expression for the average cost per month over m months.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q2/5

256

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(90 + 35m)/m

GRADING NOTE Any equivalent expression (for example 35 + 90/m).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q2/5

True or false?

An expression for a quantity that triples every 5 years, starting at 40, is $40(3)^{5t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q3/5

257

QUESTION ID
S7-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The exponent counts 5-year periods: $40(3)^{t/5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A bacteria count starts at 500 and doubles every 3 hours. Which expression gives the count after h hours?

A. $2(500)^{h/3}$

B. $500(2)^{3h}$

C. $500(3)^{h/2}$

D. $500(2)^{h/3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q4/5

258

QUESTION ID

S7-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $500(2)^{h/3}$

GRADING NOTE h/3 doubling periods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q4/5

A gym charges a $90 joining fee and $35 per month, so the average cost per month over m months is 35 + 90/m dollars. Explain what the part 90/m means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q5/5

259

QUESTION ID

S7-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The joining fee spread over m months: the part of the monthly average that comes from the fee, which shrinks as m grows.

GRADING NOTE Any equivalent explanation (the gym charges $90 to join and $35 per month).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 2

Rewrite an exponential expression to reveal a different rate • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Rewrite $12{,}500(0.97)^t$ (t in years) to show the monthly factor, rounded to 5 decimal places.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q1/5

261

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$12{,}500(0.99746)^{12t}$

GRADING NOTE Rounding to 5 decimal places required for the factor. $0.97^{1/12} \approx 0.997465$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$A(t) = 200(1/2)^{t/6}$. Write A as $200 \cdot b^t$ and give b to 4 decimal places. What is the hourly percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q2/5

262

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$b = (1/2)^{1/6} \approx 0.8909$; about 10.91% decrease per hour

GRADING NOTE Rounding to 4 decimal places for b; percent to 0.01.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q2/5

True or false?

$1.04^{n} = (1.04^{1/12})^{12n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q3/5

263

QUESTION ID

S7-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Power of a power: (1/12)(12n) = n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q3/5

Which expression shows that $500(2)^{t/3}$ grows by about 26% per time unit?

A. $1.26(500)^t$

B. $500(1.26)^{3t}$

C. $500(1.26)^t$

D. $500(2.26)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q4/5

264

QUESTION ID

S7-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $500(1.26)^t$

GRADING NOTE $2^{1/3} \approx 1.26$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q4/5

Explain why rewriting does not change the values of the function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q5/5

265

QUESTION ID

S7-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The new form is equivalent by the properties of exponents, so it gives the same output for every input; it only shows another rate.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b • SEEING STRUCTURE IN EXPRESSIONS

Interpret complicated expressions by viewing parts as single entities

MAIN	Algebra II main question	**HSA-SSE.A.1.b**
BACKWARD 1	Squares are never negative	**8.EE.A.2**
BACKWARD 2	Powers of numbers greater than 1 and between 0 and 1	**8.EE.A.1**
BACKWARD 3	A fraction gets smaller as its denominator grows	**3.NF.A.3.d**
FORWARD 1	Find the set of all values of an expression on a domain	**HSA-SSE.A.1.b**
FORWARD 2	Use single-entity reasoning to find horizontal asymptotes	**HSF-IF.C.7.d**

SET 8 • HSA-SSE.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

$P(r) = 5000(1 + r)^{10}$ with $r > 0$ is the value of an investment. Without calculating, explain why $P(r) > 5000$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

268

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

$1 + r > 1$, so $(1 + r)^{10} > 1$, and 5000 times a number greater than 1 is greater than 5000.

GRADING NOTE Any equivalent explanation that treats $(1 + r)^{10}$ as one quantity greater than 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

The height of a ball is $h(t) = 80 - 16(t - 2)^2$ feet after t seconds. Without expanding, find the greatest height and when it occurs, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

269

QUESTION ID
S8-M-Q2

TYPE
Short answer

CORRECT ANSWER
80 ft at t = 2: $(t - 2)^2 \geq 0$, so $16(t - 2)^2 \geq 0$ and $h(t) \leq 80$, with equality only when $t = 2$.

GRADING NOTE Form required: the height, the time, and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

$C(x) = 10 + 40/(x + 1)$ is a cost in dollars for $x \geq 0$. Explain why $C(x)$ is always more than 10 and never more than 50.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

270

QUESTION ID

S8-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x + 1 \geq 1$, so $40/(x + 1)$ is positive and at most 40; adding 10 gives more than 10 and at most 50 (equal to 50 at $x = 0$).

GRADING NOTE Any equivalent explanation treating $40/(x + 1)$ as one positive quantity at most 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

The temperature of a drink is $T(t) = 70 - 30(0.8)^t$ °F after t minutes. Explain why T(t) is always below 70 and gets closer to 70 as t grows.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

271

QUESTION ID

S8-M-Q4

TYPE

Short answer

CORRECT ANSWER

$30(0.8)^t$ is positive, so T is below 70; $(0.8)^t$ shrinks toward 0 as t grows, so the subtracted part shrinks and T approaches 70.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

$f(x) = 3/(x^2 + 1)$. Find the greatest value of $f(x)$ and explain why no input gives a greater value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

272

QUESTION ID

S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

3, at $x = 0$: $x^2 + 1 \geq 1$, so $3/(x^2 + 1) \leq 3$, with equality only when $x = 0$.

GRADING NOTE Form required: the value and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 1

Squares are never negative • Grade 8 • Expressions & Equations • 8.EE.A.2

True or false?

$(t - 2)^2 \geq 0$ for every real t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q1/5

274

QUESTION ID

S8-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A square of a real number is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the smallest value of $(x + 5)^2$, and for which x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q2/5

275

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0, when x = −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q2/5

Which expression can be negative for some real x?

A. $x^2 + 1$

B. $x - 1$

C. x^2

D. $(x - 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q3/5

276

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x − 1

GRADING NOTE $x - 1 < 0$ for $x < 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q3/5

What is the smallest value of $x^2 + 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q4/5

277

QUESTION ID

S8-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. At x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q4/5

True or false?

$-(x - 3)^2$ is never positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q5/5

278

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the opposite of a nonnegative number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 2

Powers of numbers greater than 1 and between 0 and 1 • Grade 8 • Expressions & Equations • 8.EE.A.1

Is 1.05^{10} greater than or less than 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q1/5

280

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Greater than 1

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q1/5

Is 0.8^5 greater than or less than 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q2/5

281

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Less than 1 (about 0.33)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As n increases, 0.5^n

A. becomes negative

B. stays 0.5

C. gets closer to 0

D. grows without bound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q3/5

282

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. gets closer to 0

GRADING NOTE Repeated halving.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q3/5

True or false?

A negative number raised to any whole-number power is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q4/5

283

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-2)^2 = 4$ is positive; even powers of negatives are positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Order from least to greatest: 0.9^2, 0.9^5, 0.9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q5/5

284

QUESTION ID

S8-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.9^5, 0.9^2, 0.9

GRADING NOTE Any equivalent listing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 3

A fraction gets smaller as its denominator grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Which is greater, 40/2 or 40/5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q1/5

286

QUESTION ID

S8-B3-Q1

TYPE

Short answer

CORRECT ANSWER

40/2

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q1/5

True or false?

If $x \geq 0$, then $40/(x + 1) \leq 40$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q2/5

287

QUESTION ID

S8-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The denominator is at least 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As x grows, 6/(x + 3)

A. increases

B. becomes negative

C. decreases toward 0

D. stays 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q3/5

288

QUESTION ID

S8-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. decreases toward 0

GRADING NOTE The denominator grows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q3/5

What is the greatest value of 12/(x + 4) for x ≥ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q4/5

289

QUESTION ID

S8-B3-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. At x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q4/5

True or false?

8/10 is greater than 8/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q5/5

290

QUESTION ID

S8-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Same numerator, larger denominator gives a smaller fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 1

Find the set of all values of an expression on a domain • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

Find every value that $h(t) = 80 - 16(t - 2)^2$ takes for $0 \le t \le 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q1/5

292

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$16 \le h \le 80$

GRADING NOTE Form required: an inequality or interval. $(t - 2)^2$ goes from 0 to 4, so $16(t - 2)^2$ from 0 to 64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find every value of $f(x) = 3/(x^2 + 1)$ for real x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q2/5

293

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$0 < f(x) \leq 3$

GRADING NOTE Form required: an inequality or interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q2/5

True or false?

For $x \geq 0$, $C(x) = 10 + 40/(x + 1)$ takes every value in $10 < C \leq 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q3/5

294

QUESTION ID

S8-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 40/(x + 1) takes every value in (0, 40].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q3/5

For $t \geq 0$, which set contains every value of $70 - 30(0.8)^t$?

A. $T \geq 70$

B. $40 \leq T < 70$

C. $40 < T \leq 70$

D. $0 < T < 70$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q4/5

295

QUESTION ID

S8-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $40 \leq T < 70$

GRADING NOTE $30(0.8)^t$ goes from 30 (at $t = 0$) toward 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q4/5

Find every value of $g(x) = 5 - 2/(x^2 + 4)$ for real x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q5/5

296

QUESTION ID

S8-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$4.5 \le g(x) < 5$

GRADING NOTE Form required: an inequality or interval. $2/(x^2 + 4)$ is in $(0, 0.5]$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 2

Use single-entity reasoning to find horizontal asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Explain why y = 10 + 40/(x + 1) has the horizontal asymptote y = 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q1/5

298

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

As x grows without bound, 40/(x + 1) gets closer to 0, so y gets closer to 10.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q1/5

Rewrite (2x + 7)/(x + 3) as 2 + 1/(x + 3) and state its horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q2/5

299

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2

GRADING NOTE Form required: the equation y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q2/5

True or false?

$y = 3/(x^2 + 1)$ has the horizontal asymptote $y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q3/5

300

QUESTION ID

S8-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3/(x^2 + 1)$ approaches 0 as $|x|$ grows; the asymptote is $y = 0$ (3 is the maximum).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q3/5

The horizontal asymptote of $y = 5 - 2/(x - 4)$ is

A. $y = -2$

B. $y = 5$

C. $y = 4$

D. $y = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q4/5

301

QUESTION ID
S8-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. y = 5

GRADING NOTE $-2/(x - 4)$ approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q4/5

Does the graph of y = 10 + 40/(x + 1) ever reach its horizontal asymptote? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q5/5

302

QUESTION ID

S8-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No: 40/(x + 1) is never 0, so y is never exactly 10.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q5/5

Factor higher-degree expressions by recognizing differences of squares and sums or differences of cubes

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Take out a greatest common factor	**6.EE.A.3**
BACKWARD 2	Recognize perfect squares and perfect cubes	**8.EE.A.2**
BACKWARD 3	Multiply to check a factoring pattern	**HSA-APR.A.1**
FORWARD 1	Factor expressions whose parts are themselves expressions	**HSA-SSE.A.2**
FORWARD 2	Complete the factorization over the complex numbers	**HSN-CN.C.8**

SET 9 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 81$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q1/5

305

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 9)(x + 3)(x - 3)$

GRADING NOTE Form required: factored completely over the integers. $(x^2)^2 - 9^2 = (x^2 - 9)(x^2 + 9)$. $x^2 + 9$ does not factor over the integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q1/5

Factor $8x^3 + 27$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q2/5

306

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 3)(4x^2 - 6x + 9)$

GRADING NOTE Form required: factored completely. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ with $a = 2x$, $b = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q2/5

Factor $x^6 - 64$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q3/5

307

QUESTION ID

S9-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x^2 + 2x + 4)(x + 2)(x^2 - 2x + 4)$

GRADING NOTE Form required: factored completely, any order. $(x^3 - 8)(x^3 + 8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q3/5

Factor $16x^4 - y^4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q4/5

308

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(4x^2 + y^2)(2x + y)(2x - y)$

GRADING NOTE Form required: factored completely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q4/5

Factor $2x^5 - 32x$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q5/5

309

QUESTION ID

S9-M-Q5

TYPE

Short answer

CORRECT ANSWER

$2x(x^2 + 4)(x + 2)(x - 2)$

GRADING NOTE Form required: factored completely. Common factor 2x first: $2x(x^4 - 16)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 1

Take out a greatest common factor • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Factor 6x + 18.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q1/5

311

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6(x + 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q1/5

Factor $2x^5 - 32x$ by taking out the greatest common factor only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q2/5

312

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$2x(x^4 - 16)$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q2/5

The greatest common factor of $12x^3$ and $18x$ is

A. $6x^3$

B. $6x$

C. $3x$

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q3/5

313

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6x

GRADING NOTE 6 and x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$3x^2 + 9x = 3x(x + 9)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q4/5

314

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3x(x + 9) = 3x^2 + 27x$; the correct factorization is $3x(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q4/5

Factor $5y^2 - 20y$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q5/5

315

QUESTION ID

S9-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5y(y − 4)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 2

Recognize perfect squares and perfect cubes • Grade 8 • Expressions & Equations • 8.EE.A.2

Write $16x^4$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q1/5

317

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(4x^2)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q1/5

Write $8x^3$ as a cube.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q2/5

318

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x)^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q2/5

Which is a perfect cube?

A. 32

B. 16

C. 36

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q3/5

319

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 64

GRADING NOTE $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q3/5

True or false?

x^6 is both a perfect square and a perfect cube.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q4/5

320

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x^3)^2$ and $(x^2)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the cube root of 27.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q5/5

321

QUESTION ID

S9-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 3

Multiply to check a factoring pattern • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(a + b)(a^2 - ab + b^2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q1/5

323

QUESTION ID

S9-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$a^3 + b^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x - 2)(x^2 + 2x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q2/5

324

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 - 8$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q2/5

True or false?

$(x + 3)(x^2 - 6x + 9) = x^3 + 27.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q3/5

325

QUESTION ID

S9-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The middle term should be $-3x$: $(x + 3)(x^2 - 3x + 9) = x^3 + 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x^2 - 9)(x^2 + 9)$ equals

A. $x^4 - 18x^2 - 81$

B. $x^4 + 81$

C. $(x - 3)^4$

D. $x^4 - 81$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q4/5

326

QUESTION ID

S9-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x^4 - 81$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q4/5

Multiply $(2x + 1)(4x^2 - 2x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q5/5

327

QUESTION ID

S9-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$8x^3 + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 1

Factor expressions whose parts are themselves expressions • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $(x + 1)^2 - 9$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q1/5

329

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 4)$

GRADING NOTE Form required: factored completely. $((x + 1) - 3)((x + 1) + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $(2x - 1)^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q2/5

330

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 1)(4x^2 - 8x + 7)$

GRADING NOTE Form required: factored completely. a = 2x − 1, b = 2: $(a + b)(a^2 - 2a + 4) = (2x + 1)((2x - 1)^2 - 2(2x - 1) + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q2/5

True or false?

$a^6 - b^6$ can be factored first as a difference of squares or first as a difference of cubes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q3/5

331

QUESTION ID

S9-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(a^3)^2 - (b^3)^2$ or $(a^2)^3 - (b^2)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q3/5

$(x - y)^2 - (x + y)^2$ equals

A. 0

B. $4xy$

C. $-4xy$

D. $-2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q4/5

332

QUESTION ID

S9-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-4xy$

GRADING NOTE $((x - y) - (x + y))((x - y) + (x + y)) = (-2y)(2x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q4/5

Factor $x^2(x + 2) - 4(x + 2)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q5/5

333

QUESTION ID

S9-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + 2)^2(x - 2)$

GRADING NOTE Form required: factored completely. $(x + 2)(x^2 - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 2

Complete the factorization over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^4 - 81$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q1/5

335

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(x - 3)(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $16x^4 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q2/5

336

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 1)(2x - 1)(2x + i)(2x - i)$

GRADING NOTE Any equivalent factorization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q2/5

True or false?

Over the complex numbers, $x^3 + 8$ has only one linear factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q3/5

337

QUESTION ID

S9-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has three: $(x + 2)$ and two from $x^2 - 2x + 4 = (x - 1 - i\sqrt{3})(x - 1 + i\sqrt{3})$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q3/5

Which polynomial is fully factored over the integers but not over the complex numbers?

A. $x^2 + 9$

B. $x - 9$

C. $x^2 - 9$

D. $(x - 3)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q4/5

338

QUESTION ID

S9-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 9$

GRADING NOTE $x^2 + 9 = (x + 3i)(x - 3i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q4/5

$2x^5 - 32x = 2x(x^2 + 4)(x + 2)(x - 2)$ over the integers. Complete the factorization over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q5/5

339

QUESTION ID

S9-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2x(x + 2i)(x - 2i)(x + 2)(x - 2)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q5/5

SET 10 • HSA-SSE.A.2 • SEEING STRUCTURE IN EXPRESSIONS

Rewrite expressions in quadratic form and factor

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Factor quadratic trinomials	**HSA-SSE.A.2**
BACKWARD 2	Write powers as squares of other powers	**8.EE.A.1**
BACKWARD 3	Treat a repeated part as a single variable	**HSA-SSE.A.1.b**
FORWARD 1	Solve equations in quadratic form over the complex numbers	**HSN-CN.C.7**
FORWARD 2	Solve exponential equations in quadratic form	**HSF-LE.A.4**

SET 10 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 5x^2 + 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q1/5

342

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1)(x - 2)(x + 2)$

GRADING NOTE Form required: factored completely. With $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q1/5

Factor $x^4 + 3x^2 - 10$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q2/5

343

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 5)(x^2 - 2)$

GRADING NOTE Form required: factored completely over the integers ($x^2 - 2$ does not factor over the integers). $u^2 + 3u - 10 = (u + 5)(u - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q2/5

Factor $9^x - 4 \cdot 3^x + 3$ by writing it in quadratic form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q3/5

344

QUESTION ID

S10-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(3^x - 1)(3^x - 3)$

GRADING NOTE Form required: a product of two factors. $9^x = (3^x)^2$, so with $u = 3^x$: $u^2 - 4u + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q3/5

Factor $x^6 - 9x^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q4/5

345

QUESTION ID

S10-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x^2 + x + 1)(x - 2)(x^2 + 2x + 4)$

GRADING NOTE Form required: factored completely. $u = x^3$: $(u - 1)(u - 8)$; then difference of cubes twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q4/5

Factor $2x^4 - 7x^2 - 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q5/5

346

QUESTION ID

S10-M-Q5

TYPE

Short answer

CORRECT ANSWER

$(2x^2 + 1)(x - 2)(x + 2)$

GRADING NOTE Form required: factored completely. $2u^2 - 7u - 4 = (2u + 1)(u - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 1

Factor quadratic trinomials • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $u^2 - 5u + 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q1/5

348

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(u − 1)(u − 4)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B1-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q1/5

Factor $u^2 + 3u - 10$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q2/5

349

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(u + 5)(u − 2)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B1-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q2/5

$2u^2 - 7u - 4$ equals

A. $(2u + 1)(u - 4)$

B. $(u - 2)(2u + 2)$

C. $(2u + 4)(u - 1)$

D. $(2u - 1)(u + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q3/5

350

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2u + 1)(u − 4)

GRADING NOTE Check: $2u^2 - 8u + u - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$u^2 - 9u + 8 = (u + 1)(u - 8)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q4/5

351

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(u + 1)(u - 8) = u^2 - 7u - 8$; the correct factorization is $(u - 1)(u - 8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $u^2 - 4u + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q5/5

352

QUESTION ID

S10-B1-Q5

TYPE

Short answer

CORRECT ANSWER

(u − 1)(u − 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 2

Write powers as squares of other powers • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write x^4 as a power of x^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

354

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

Write 9^x as a power of 3^x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

355

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(3^x)^2$

GRADING NOTE Any equivalent form. $9^x = (3^2)^x = (3^x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

x^6 equals

A. $(x^4)^2$

B. $(x^3)^2$

C. $x^3 \cdot x^3 \cdot x^3$

D. $(x^2)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

356

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x^3)^2$

GRADING NOTE $3 \cdot 2 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$4^x = (2^x)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

357

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^x = 2^{2x}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

Write 25^x as a power of 5^x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

358

QUESTION ID

S10-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(5^x)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 3

Treat a repeated part as a single variable • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

In $x^4 - 5x^2 + 4$, what single variable u turns the expression into $u^2 - 5u + 4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q1/5

360

QUESTION ID

S10-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$u = x^2$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $2(x - 1)^2 + 3(x - 1) - 5$ using $u = x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q2/5

361

QUESTION ID

S10-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$2u^2 + 3u - 5$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q2/5

True or false?

In $4^x - 2^x - 6$, the substitution $u = 2^x$ gives $u^2 - u - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q3/5

362

QUESTION ID

S10-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^x = (2^x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q3/5

Which substitution writes $x^6 - 9x^3 + 8$ as a quadratic?

A. $u = x^2$

B. $u = x^3$

C. $u = x^6$

D. $u = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q4/5

363

QUESTION ID

S10-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $u = x^3$

GRADING NOTE $x^6 = (x^3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q4/5

After factoring $(u - 1)(u - 8)$ with $u = x^3$, write the result in terms of x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q5/5

364

QUESTION ID

S10-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$(x^3 - 1)(x^3 - 8)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 1

Solve equations in quadratic form over the complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^4 + 3x^2 - 4 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q1/5

366

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 1, -1, 2i, -2i$

GRADING NOTE Form required: all four. $x^2 = 1$ or $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q1/5

Solve $x^4 - 5x^2 + 4 = 0$ over the complex numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q2/5

367

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = ±1, ±2

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q2/5

True or false?

$x^4 + 10x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q3/5

368

QUESTION ID

S10-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x^2 + 1)(x^2 + 9) = 0$ has only the solutions $\pm i$ and $\pm 3i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q3/5

How many complex solutions, with multiplicity, does $x^4 - 13x^2 + 36 = 0$ have?

A. 1

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q4/5

369

QUESTION ID

S10-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE ±2 and ±3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q4/5

Solve $x^4 + x^2 - 12 = 0$ over the complex numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q5/5

370

QUESTION ID

S10-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x = \pm\sqrt{3}, \pm 2i$

GRADING NOTE Form required: all four, exact. $(x^2 + 4)(x^2 - 3) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 2

Solve exponential equations in quadratic form • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $9^x - 5 \cdot 3^x + 6 = 0$. Give exact solutions.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q1/5

372

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \log_3 2$ or $x = 1$

GRADING NOTE Form required: both, exact. $3^x = 2$ or $3^x = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q1**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q1/5

Solve $4^x - 6 \cdot 2^x + 8 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q2/5

373

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 1 or x = 2

GRADING NOTE Form required: both. $2^x = 2$ or $2^x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q2/5

True or false?

$4^x + 2^x - 2 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q3/5

374

QUESTION ID

S10-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^x = 1$ gives $x = 0$; $2^x = -2$ is impossible since $2^x > 0$. One solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solutions of $25^x - 26 \cdot 5^x + 25 = 0$ are

A. 0 and 2

B. 1 and 25

C. 0 and 1

D. 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q4/5

375

QUESTION ID
S10-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 0 and 2

GRADING NOTE $5^x = 1$ or $5^x = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q4/5

Solve $e^{2t} - 7e^{t} + 10 = 0$. Give exact solutions and decimal values to 3 decimal places.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q5/5

376

QUESTION ID

S10-F2-Q5

TYPE

Short answer

CORRECT ANSWER

t = ln 2 ≈ 0.693 or
t = ln 5 ≈ 1.609

GRADING NOTE Form required: exact and 3 decimal places. $e^{t} = 2$ or $e^{t} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4 • SEEING STRUCTURE IN EXPRESSIONS

Derive the formula for the sum of a finite geometric series

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Write terms of a geometric sequence	**HSF-LE.A.2**
BACKWARD 2	Distribute a factor over a sum	**6.EE.A.3**
BACKWARD 3	Use the product rule for exponents	**8.EE.A.1**
FORWARD 1	Prove the sum formula from a polynomial identity	**HSA-APR.C.4**
FORWARD 2	Describe how partial sums behave when the ratio is between -1 a...	**HSA-SSE.B.4**

SET 11 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Let $S_n = a + ar + ar^2 + \ldots + ar^{n-1}$ with $r \neq 1$. Derive $S_n = a(1 - r^n)/(1 - r)$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q1/5

379

QUESTION ID

S11-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$S_n - rS_n = a - ar^n$, so $(1 - r)S_n = a(1 - r^n)$ and $S_n = a(1 - r^n)/(1 - r)$.

GRADING NOTE

REQUIRED WORK: Write rS_n, subtract it from S_n, and divide by $1 - r$.

Any valid proof or derivation. Essential: $rS_n = ar + ar^2 + \ldots + ar^n$; all middle terms cancel in $S_n - rS_n$; factor $(1 - r)$; divide, which needs $r \neq 1$. Invalid: checking the formula for one or two values of n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the multiply-and-subtract method (not the formula) to find $S = 3 + 6 + 12 + \ldots + 3 \cdot 2^9$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q2/5

380

QUESTION ID

S11-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

S = 3069

GRADING NOTE REQUIRED WORK: Write 2S, subtract S from 2S.

Any valid proof or derivation. Essential: $2S = 6 + 12 + \ldots + 3 \cdot 2^{10}$; $2S - S = 3 \cdot 2^{10} - 3 = 3072 - 3 = 3069$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q2/5

Explain why the derivation of $S_n = a(1 - r^n)/(1 - r)$ does not work when $r = 1$, and find S_n in that case.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q3/5

381

QUESTION ID

S11-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The last step divides by $1 - r = 0$; when $r = 1$ every term is a, so $S_n = na$.

GRADING NOTE

REQUIRED WORK: Look at the division step and at the series itself when r = 1.

Any valid explanation. Essential: division by zero in the derivation; the n equal terms add to na. Invalid: plugging r = 1 into the formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q3/5

Derive a formula for $T = 1 + x + x^2 + \ldots + x^n$ ($n + 1$ terms, $x \neq 1$), then check it for $x = 2$ and $n = 3$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q4/5

382

QUESTION ID

S11-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$T = (x^{n+1} - 1)/(x - 1)$; for $x = 2$, $n = 3$: $(16 - 1)/1 = 15 = 1 + 2 + 4 + 8$

GRADING NOTE REQUIRED WORK: Multiply by x and subtract.

Any valid proof or derivation. Essential: $xT = x + \ldots + x^{n+1}$; $xT - T = x^{n+1} - 1$; divide by $x - 1$ ($x \neq 1$); the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q4/5

Derive the total of 20 yearly amounts $100, 100(1.05), 100(1.05)^2, \ldots, 100(1.05)^{19}$ by the multiply-and-subtract method, and give it to the nearest cent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q5/5

383

QUESTION ID

S11-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$T = 100(1.05^{20} - 1)/0.05 \approx \$3306.60$

GRADING NOTE

REQUIRED WORK: Write 1.05T and subtract T.

Any valid proof or derivation. Essential: $1.05T - T = 100(1.05)^{20} - 100$; $0.05T = 100(1.05^{20} - 1)$; $T \approx 3306.60$. Rounding to the nearest cent required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 1

Write terms of a geometric sequence • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the first four terms of the geometric sequence with first term 3 and ratio 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q1/5

385

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3, 6, 12, 24

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the 10th term of 3, 6, 12, ...?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q2/5

386

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$3 \cdot 2^9 = 1536$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q2/5

The nth term of a geometric sequence with first term a and ratio r is

A. $a + (n - 1)r$

B. $a \cdot n \cdot r$

C. ar^n

D. ar^{n-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q3/5

387

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ar^{n-1}

GRADING NOTE The first term is ar^0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q3/5

True or false?

100, 105, 110.25, ... is arithmetic with common difference 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q4/5

388

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences are 5 and 5.25; the ratio 1.05 is constant, so it is geometric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the ratio of 81, 27, 9, 3, ...

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q5/5

389

QUESTION ID

S11-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 2

Distribute a factor over a sum • Grade 6 • Expressions & Equations • 6.EE.A.3

Distribute: 2(3 + 6 + 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q1/5

391

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 + 12 + 24

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distribute: $r(a + ar + ar^2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q2/5

392

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$ar + ar^2 + ar^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q2/5

True or false?

$x(1 + x + x^2) = x + x^2 + x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q3/5

393

QUESTION ID

S11-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distributive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q3/5

1.05(100 + 105) equals

A. 105 + 105

B. 100 + 110.25

C. 105 + 110.25

D. 205

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q4/5

394

QUESTION ID

S11-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 105 + 110.25

GRADING NOTE Each term is multiplied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q4/5

Subtract: (6 + 12 + 24 + 48) − (3 + 6 + 12 + 24).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q5/5

395

QUESTION ID

S11-B2-Q5

TYPE

Short answer

CORRECT ANSWER

45 (= 48 − 3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 3

Use the product rule for exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $r \cdot r^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q1/5

397

QUESTION ID

S11-B3-Q1

TYPE

Short answer

CORRECT ANSWER

r^5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $ar^{n-1} \cdot r$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q2/5

398

QUESTION ID

S11-B3-Q2

TYPE

Short answer

CORRECT ANSWER

ar^n

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$2 \cdot 2^9$ equals

A. 2^{10}

B. 4^{10}

C. 4^9

D. 2^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q3/5

399

QUESTION ID

S11-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2^{10}

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q3/5

True or false?

$x \cdot x^n = x^{2n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q4/5

400

QUESTION ID

S11-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Add exponents: $x^1 \cdot x^n = x^{n+1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q4/5

Evaluate 2^{10}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q5/5

401

QUESTION ID

S11-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1024

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 1

Prove the sum formula from a polynomial identity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Prove the identity

$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \ldots + x + 1)$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q1/5

403

QUESTION ID

S11-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The product telescopes to $x^n - 1$.

GRADING NOTE REQUIRED WORK: Expand the right side and cancel.

Any valid proof or derivation. Essential: x times the sum gives $x^n + \ldots + x$; minus the sum gives $-(x^{n-1} + \ldots + 1)$; all but x^n and -1 cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q1/5

Use the identity $x^n - 1 = (x - 1)(x^{n-1} + \ldots + 1)$ to write $1 + 3 + 9 + \ldots + 3^5$ as a quotient, and evaluate it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q2/5

404

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(3^6 - 1)/(3 - 1)$
$= 728/2 = 364$

GRADING NOTE Form required: the quotient and the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q2/5

True or false?

The identity shows that $x - 1$ is a factor of $x^n - 1$ for every whole number $n \geq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q3/5

405

QUESTION ID

S11-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It writes $x^n - 1$ as $(x - 1)$ times a polynomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q3/5

Which equals $1 + 2 + 4 + \ldots + 2^9$?

A. 2^{10}

B. $2^{10} - 1$

C. $2^9 - 1$

D. $(2^9 - 1)/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q4/5

406

QUESTION ID

S11-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $2^{10} - 1$

GRADING NOTE $(2^{10} - 1)/(2 - 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q4/5

Explain how the identity gives $a(1 - r^n)/(1 - r)$ for $a + ar + \ldots + ar^{n-1}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q5/5

407

QUESTION ID

S11-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Put x = r and multiply by a: $a(r^n - 1) = a(r - 1)(1 + r + \ldots + r^{n-1})$; divide by r − 1 (r ≠ 1) and rewrite signs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 2

Describe how partial sums behave when the ratio is between −1 and 1 • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

For the series 8 + 4 + 2 + 1 + ..., find S_5 and S_{10} to 4 decimal places.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q1/5

409

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$S_5 = 15.5000$;

$S_{10} = 15.9844$

GRADING NOTE Rounding to 4 decimal places. $S_n = 8(1 - 0.5^n)/0.5 = 16(1 - 0.5^n)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q1/5

Using $S_n = 16(1 - 0.5^n)$, explain why every partial sum of $8 + 4 + 2 + \ldots$ is less than 16 but can be made as close to 16 as you like.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q2/5

410

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.5^n is positive, so $S_n < 16$; 0.5^n shrinks toward 0 as n grows, so S_n gets as close to 16 as you like.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q2/5

True or false?

For r = 2, the partial sums of $a + ar + ar^2 + \ldots$ ($a > 0$) stay below a fixed number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q3/5

411

QUESTION ID

S11-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE r^n grows without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q3/5

For 10 − 5 + 2.5 − 1.25 + …, the partial sums get close to

A. 20/3

B. 15

C. 5

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q4/5

412

QUESTION ID

S11-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 20/3

GRADING NOTE a/(1 − r) = 10/1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q4/5

A ball travels 10 m down, then rebounds and falls in a geometric pattern so the total distance after n bounces is $10 + 2 \cdot 6(1 - 0.6^n)/0.4$. What total can the distance never reach, and why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q5/5

413

QUESTION ID

S11-F2-Q5

TYPE

Short answer

CORRECT ANSWER

40 m: $2 \cdot 6/0.4 = 30$ and $10 + 30 = 40$; $(1 - 0.6^n) < 1$ always.

GRADING NOTE Form required: the value and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q5/5

Use the finite geometric series formula to solve problems

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Identify the first term, ratio, and number of terms	**HSF-LE.A.2**
BACKWARD 2	Change a percent to a growth factor	**7.RP.A.3**
BACKWARD 3	Round results of calculations correctly	**HSN-Q.A.3**
FORWARD 1	Find the number of terms with logarithms	**HSF-LE.A.4**
FORWARD 2	Compare two payment plans with the series formula	**HSA-SSE.B.4**

SET 12 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 12 terms of 5 + 15 + 45 + ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q1/5

416

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

1,328,600

GRADING NOTE Any equivalent form. $5(3^{12} - 1)/(3 - 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q1/5

Leah deposits $250 at the end of each month into an account that earns 0.5% interest per month. The balance just after the 36th deposit is $250 + 250(1.005) + \ldots + 250(1.005)^{35}$. Find it to the nearest cent.

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q2/5

417

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

$9834.03

GRADING NOTE Rounding to the nearest cent required. $250(1.005^{36} - 1)/0.005 = 9834.03$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q2/5

A $180,000 loan is repaid with 360 equal monthly payments P at a monthly interest rate of 0.4%. P satisfies
$180{,}000 = P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$. Use the geometric series formula to find P to the nearest cent.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q3/5

QUESTION ID
S12-M-Q3

TYPE
Short answer

CORRECT ANSWER
$944.40

GRADING NOTE Rounding to the nearest cent required. a = P/1.004, r = 1/1.004, n = 360: P = 180,000 · $0.004/(1 - 1.004^{-360}) = 944.40$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q3/5

A ball is dropped from 10 m and each rebound reaches 60% of the previous height. Find the total vertical distance it has traveled when it hits the ground for the 6th time, to the nearest hundredth of a meter.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q4/5

419

QUESTION ID

S12-M-Q4

TYPE

Short answer

CORRECT ANSWER

37.67 m

GRADING NOTE Rounding to 0.01 required. $10 + 2(6 + 3.6 + 2.16 + 1.296 + 0.7776) = 10 + 2 \cdot 6(1 - 0.6^5)/0.4 = 37.6672$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q4/5

What is the least number of terms of 2 + 6 + 18 + ... whose sum is greater than 10,000? Show the sums that decide it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q5/5

420

QUESTION ID

S12-M-Q5

TYPE

Short answer

CORRECT ANSWER

9 terms: $S_8 = 6560$ and $S_9 = 19{,}682$

GRADING NOTE Form required: n and the two sums. $S_n = 3^n - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 1

Identify the first term, ratio, and number of terms • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

For 5 + 15 + 45 + ... (12 terms), give a, r, and n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q1/5

422

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

a = 5, r = 3, n = 12

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q1/5

For $250 + 250(1.005) + \ldots + 250(1.005)^{35}$, give a, r, and n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q2/5

423

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 250, r = 1.005, n = 36

GRADING NOTE Form required: all three. Exponents 0 to 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q2/5

For 6 + 3.6 + 2.16 + 1.296 + 0.7776, r is

A. 0.4

B. 1.67

C. 0.6

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q3/5

424

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.6

GRADING NOTE 3.6/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q3/5

True or false?

A list with exponents 0 through 35 has 35 terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q4/5

425

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 36 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q4/5

For $P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$, give the first term and the ratio.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q5/5

426

QUESTION ID

S12-B1-Q5

TYPE

Short answer

CORRECT ANSWER

a = P/1.004,

r = 1/1.004

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 2

Change a percent to a growth factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

What factor adds 0.5% interest?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q1/5

428

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.005

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A height is 60% of the one before. What is the ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q2/5

429

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

0.4% per month as a factor is

A. 1.04

B. 1.004

C. 0.004

D. 1.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q3/5

430

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.004

GRADING NOTE 1 + 0.004.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q3/5

True or false?

Interest of 12% per year is a factor of 0.12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q4/5

431

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The growth factor is 1 + 0.12 = 1.12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A loss of 15% per year is what factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q5/5

QUESTION ID

S12-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.85

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 3

Round results of calculations correctly • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Round 9834.0262 to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q1/5

434

QUESTION ID

S12-B3-Q1

TYPE

Short answer

CORRECT ANSWER

9834.03

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 37.6672 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q2/5

435

QUESTION ID

S12-B3-Q2

TYPE

Short answer

CORRECT ANSWER

37.67

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q2/5

True or false?

Rounding 1.005 to 1.01 before raising it to the 36th power gives the same answer to the nearest cent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q3/5

436

QUESTION ID

S12-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Early rounding changes the result a lot; round only at the end.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q3/5

1.005^{36} is closest to

A. 1.1967

B. 1.18

C. 1.2

D. 1.036

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q4/5

437

QUESTION ID

S12-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1.1967

GRADING NOTE Calculator value 1.19668....

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 3^{12} exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q5/5

438

QUESTION ID

S12-B3-Q5

TYPE

Short answer

CORRECT ANSWER

531441

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 1

Find the number of terms with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

How many end-of-month deposits of $250 at 0.5% per month are needed for the balance to reach $20,000? Use $250(1.005^n - 1)/0.005 \geq 20{,}000$.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q1/5

440

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

68 deposits

GRADING NOTE Form required: a whole number. $1.005^n \geq 1 + 20{,}000(0.005)/250 = 1.4$, so $n \geq \ln 1.4/\ln 1.005 \approx 67.46$; check: n = 67 gives about $19,838.70 and n = 68 about $20,187.89.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q1/5

Solve $2(3^n - 1)/2 = 3^n - 1 = 1{,}000{,}000$ for n exactly, then give the least whole number of terms whose sum is at least 1,000,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q2/5

441

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$n = \log_3(1{,}000{,}001)$

≈ 12.58, so 13 terms

GRADING NOTE Form required: the exact value and the whole number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q2/5

True or false?

For a ratio greater than 1, more terms always give a larger sum when a > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q3/5

442

QUESTION ID
S12-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each added term is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q3/5

The sum $5(2^n - 1)$ first exceeds 5,000 when n =

A. 9

B. 11

C. 10

D. 1000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q4/5

443

QUESTION ID

S12-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE $2^n > 1001$; $2^{10} = 1024$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q4/5

Explain why the number of terms must be rounded up, not to the nearest whole number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q5/5

444

QUESTION ID

S12-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The sum must reach the target; rounding down would give a sum that is still too small.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 2

Compare two payment plans with the series formula • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Plan A pays $1,000 per month for 24 months. Plan B pays $500 the first month and 8% more each month after that, for 24 months. Which plan pays more in total, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q1/5

446

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Plan B: $500(1.08^{24} - 1)/0.08 \approx \$33{,}382.38$ vs $24,000; about $9,382.38 more

GRADING NOTE Form required: the plan and the difference to the nearest cent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q1/5

Plan A pays $1,000 per month. Plan B pays $500 the first month and 8% more each month after that. After how many months does Plan B's total first exceed Plan A's total for the same number of months?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q2/5

447

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

After 18 months

GRADING NOTE Form required: the month. Compare $500(1.08^n - 1)/0.08$ with $1000n$: $n = 17$ gives $16{,}875.11 < 17{,}000$; $n = 18$ gives $18{,}725.12 > 18{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q2/5

True or false?

A geometric plan with ratio greater than 1 eventually pays more in total than any plan with a fixed monthly amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q3/5

448

QUESTION ID
S12-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its terms grow without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q3/5

Plan C pays $2,000 then 5% less each month. Its total over 24 months is closest to

A. $40,000

B. $24,000

C. $48,000

D. $28,320

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q4/5

449

QUESTION ID

S12-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $28,320

GRADING NOTE $2000(1 - 0.95^{24})/0.05 \approx 28{,}320$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q4/5

Name one assumption in comparing the plans by their totals only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q5/5

450

QUESTION ID
S12-F2-Q5

TYPE
Short answer

CORRECT ANSWER
For example: money received later is treated as equal in value to money now (no interest or inflation).

GRADING NOTE Any reasonable assumption.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q5/5

Add and subtract polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Combine like terms with the same power	**7.EE.A.1**
BACKWARD 2	Subtract a whole expression	**6.EE.A.3**
BACKWARD 3	Write polynomials in standard form	**HSA-SSE.A.1.a**
FORWARD 1	Reason about the degree of a sum or difference	**HSA-APR.A.1**
FORWARD 2	Add and subtract rational expressions	**HSA-APR.D.7**

SET 13 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Simplify $(4x^3 - 2x^2 + 7) + (-x^3 + 5x - 9)$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q1/5

453

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^3 - 2x^2 + 5x - 2$

GRADING NOTE Form required: standard form (descending powers). $4x^3 - x^3 = 3x^3$; $-2x^2$; $5x$; $7 - 9 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q1/5

Simplify $(6x^4 - x^3 + 2x) - (2x^4 + 3x^3 - x^2 + 2x - 1)$.
Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q2/5

454

QUESTION ID
S13-M-Q2

TYPE
Short answer

CORRECT ANSWER
$4x^4 - 4x^3 + x^2 + 1$

GRADING NOTE Form required: standard form. The x terms cancel: $2x - 2x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q2/5

A company's revenue is $R(x) = -0.01x^3 + 3x^2 + 40x$ dollars and its cost is $C(x) = 0.02x^2 + 15x + 900$ dollars for x items. Write the profit $P(x) = R(x) - C(x)$ in standard form.

$P(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q3/5

455

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

$-0.01x^3 + 2.98x^2 + 25x - 900$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q3/5

Simplify $(x^5 - 3x^3 + x) - (x^5 + 2x^4 - 3x^3 - 7)$. Write the result in standard form and give its degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q4/5

456

QUESTION ID

S13-M-Q4

TYPE

Short answer

CORRECT ANSWER

$-2x^4 + x + 7$; degree 4

GRADING NOTE Form required: standard form and the degree. The x^5 and x^3 terms cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q4/5

$p(x) = 5x^3 + x^2 - 6$. Find the polynomial $q(x)$ for which $p(x) + q(x) = 2x^3 - x + 4$.

$q(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q5/5

457

QUESTION ID

S13-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-3x^3 - x^2 - x + 10$

GRADING NOTE Form required: standard form. $q = (2x^3 - x + 4) - (5x^3 + x^2 - 6)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 1

Combine like terms with the same power • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $4x^3 - x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

459

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $-2x^2 + 5x - 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

460

QUESTION ID

S13-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-5x^2 + 5x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

True or false?

$x^3 + x^2 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

461

QUESTION ID

S13-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x^3 and x^2 are unlike terms; they cannot be combined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

Which pair are like terms?

A. $4x^2$ and $-x^2$

B. $4x^2$ and $4x$

C. x^3 and $3x$

D. x and 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

462

QUESTION ID

S13-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $4x^2$ and $-x^2$

GRADING NOTE Same variable, same power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $7 - 9 + 2x - 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

463

QUESTION ID

S13-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 2

Subtract a whole expression • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $-(2x^4 + 3x^3 - x^2 + 2x - 1)$ without parentheses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q1/5

465

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$-2x^4 - 3x^3 + x^2 - 2x + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q1/5

Simplify (5 − x) − (3 − 4x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q2/5

466

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2 + 3x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q2/5

True or false?

$a - (b - c) = a - b - c.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q3/5

467

QUESTION ID

S13-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $a - (b - c) = a - b + c.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q3/5

$-(x^2 - 7)$ equals

A. $x^2 + 7$

B. $-x^2 - 7$

C. $-x^2 + 7$

D. $x^2 - 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q4/5

468

QUESTION ID

S13-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-x^2 + 7$

GRADING NOTE Distribute -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify 10 − (x + 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q5/5

469

QUESTION ID

S13-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6 − x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 3

Write polynomials in standard form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Write $7 + 5x - 2x^3 + x^2$ in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q1/5

471

QUESTION ID

S13-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$-2x^3 + x^2 + 5x + 7$

GRADING NOTE Form required: descending powers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q1/5

What is the leading coefficient of $3 - 4x^2 + x^4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q2/5

472

QUESTION ID

S13-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q2/5

The degree of $-2x^4 + x + 7$ is

A. 7

B. 1

C. 4

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q3/5

473

QUESTION ID

S13-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE Highest power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q3/5

True or false?

The standard form of $2x - x^3$ is $2x - x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q4/5

474

QUESTION ID

S13-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Standard form lists powers in descending order: $-x^3 + 2x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q4/5

What is the constant term of $6x^3 - 2x + 9$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q5/5

475

QUESTION ID

S13-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 1

Reason about the degree of a sum or difference • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

p(x) has degree 4 and q(x) has degree 4. What degrees can p(x) + q(x) have? Give an example where the degree drops.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q1/5

477

QUESTION ID
S13-F1-Q1

TYPE
Short answer

CORRECT ANSWER
4 or less (or the zero polynomial); for example $(x^4 + x) + (-x^4 + 1) = x + 1$

GRADING NOTE Any correct example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q1**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q1/5

For which value of k does
$(kx^3 + 2x - 1) - (5x^3 - x^2 + 3)$ have degree 2?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q2/5

478

QUESTION ID
S13-F1-Q2

TYPE
Short answer

CORRECT ANSWER
5

GRADING NOTE Any equivalent form. The x^3 terms cancel when k = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q2/5

True or false?

The difference of two polynomials is always a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q3/5

479

QUESTION ID
S13-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Polynomials are closed under subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q3/5

p has degree 5 and q has degree 3. The degree of $p - q$ is

A. 5

B. 2

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q4/5

480

QUESTION ID
S13-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 5

GRADING NOTE The x^5 term of p cannot cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q4**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q4/5

Explain why the sum of two polynomials is a polynomial.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q5/5

481

QUESTION ID
S13-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Adding like terms gives terms that are a coefficient times a whole-number power of x, which is the form of a polynomial.

GRADING NOTE Any equivalent explanation (closure under addition).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 2

Add and subtract rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Write 3/x + 2/(x + 1) as one rational expression, and state the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q1/5

483

QUESTION ID
S13-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(5x + 3)/(x(x + 1)); x ≠ 0, −1

GRADING NOTE Form required: one fraction and the exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q1/5

Write 1/(x − 2) − 1/(x + 2) as one rational expression, and state the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q2/5

484

QUESTION ID
S13-F2-Q2

TYPE
Short answer

CORRECT ANSWER
4/((x − 2)(x + 2)); x ≠ 2, −2

GRADING NOTE Form required: one fraction and the exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q2**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q2/5

True or false?

Polynomials are closed under division.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q3/5

485

QUESTION ID
S13-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For example 1 ÷ x is not a polynomial; rational expressions are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

x/2 + x/3 equals

A. 5x/5

B. x/5

C. 5x/6

D. 2x/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q4/5

486

QUESTION ID
S13-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 5x/6

GRADING NOTE Common denominator 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q4/5

Explain how adding rational expressions is like adding fractions of integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q5/5

487

QUESTION ID
S13-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Rewrite both over a common denominator, then add the numerators; polynomials play the role of integers.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q5/5

Multiply polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Distribute a monomial over a polynomial	**HSA-APR.A.1**
BACKWARD 2	Multiply powers with the same base	**8.EE.A.1**
BACKWARD 3	Multiply two binomials	**HSA-APR.A.1**
FORWARD 1	Find a coefficient of a product without full expansion	**HSA-APR.A.1**
FORWARD 2	Expand powers of binomials with the Binomial Theorem	**HSA-APR.C.5**

SET 14 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(x^2 + 3x - 2)(2x - 5)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q1/5

490

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

$2x^3 + x^2 - 19x + 10$

GRADING NOTE Form required: standard form. $2x^3 - 5x^2 + 6x^2 - 15x - 4x + 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q1/5

Expand $(x - 2)^3$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q2/5

491

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 - 6x^2 + 12x - 8$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q2/5

Multiply $(x^2 - x + 1)(x^2 + x + 1)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q3/5

492

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x^4 + x^2 + 1$

GRADING NOTE Form required: standard form. $((x^2 + 1) - x)((x^2 + 1) + x) = (x^2 + 1)^2 - x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q3/5

Multiply (2x + 1)(x − 3)(x + 4). Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q4/5

493

QUESTION ID

S14-M-Q4

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 3x^2 - 23x - 12$

GRADING NOTE Form required: standard form. $(x - 3)(x + 4) = x^2 + x - 12$; times $(2x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q4/5

A box has length x + 2, width x − 1, and height 2x + 3 (in cm). Write its volume as a polynomial in standard form.

V(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q5/5

494

QUESTION ID

S14-M-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 5x^2 - x - 6$

GRADING NOTE Form required: standard form. $(x^2 + x - 2)(2x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 1

Distribute a monomial over a polynomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $2x(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q1/5

496

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 6x^2 - 4x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q1/5

Multiply $-5(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q2/5

497

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-5x^2 - 15x + 10$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q2/5

True or false?

$x^2(x + 4) = x^3 + 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q3/5

498

QUESTION ID

S14-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 \cdot 4 = 4x^2$, so the product is $x^3 + 4x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q3/5

$-3x(x - 1)$ equals

A. $3x^2 - 3x$

B. $-3x^2 + 3x$

C. $-3x^2 - 3x$

D. $-3x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q4/5

499

QUESTION ID

S14-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $-3x^2 + 3x$

GRADING NOTE $-3x \cdot -1 = 3x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $x(x^2 - x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q5/5

500

QUESTION ID

S14-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^3 - x^2 + x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 2

Multiply powers with the same base • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $x^2 \cdot x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q1/5

502

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x^5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $2x \cdot 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q2/5

503

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$6x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q2/5

True or false?

$x^2 \cdot x^2 = x^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q3/5

504

QUESTION ID

S14-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$-x \cdot x^3$ equals

A. x^4

B. $-x^3$

C. $-x^4$

D. $-x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q4/5

505

QUESTION ID

S14-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-x^4$

GRADING NOTE $-1 \cdot x^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(x^2)(-4x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q5/5

506

QUESTION ID

S14-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$-4x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 3

Multiply two binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x - 3)(x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q1/5

508

QUESTION ID

S14-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + x - 12$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x + 2)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q2/5

509

QUESTION ID

S14-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + x - 2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - 2)^2$ equals

A. $x^2 + 4$

B. $x^2 - 4$

C. $x^2 - 4x + 4$

D. $x^2 - 2x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q3/5

510

QUESTION ID

S14-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2 - 4x + 4$

GRADING NOTE Square of a binomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q3/5

True or false?

$(2x + 1)(x - 3) = 2x^2 - 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q4/5

511

QUESTION ID

S14-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The middle terms $-6x + x = -5x$ are missing: $2x^2 - 5x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q4/5

Multiply $(2x + 3)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q5/5

512

QUESTION ID

S14-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2 + x - 3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 1

Find a coefficient of a product without full expansion • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Find the coefficient of x^2 in $(x^2 + 3x - 2)(4x^2 - x + 5)$ without expanding everything.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q1/5

514

QUESTION ID
S14-F1-Q1

TYPE
Short answer

CORRECT ANSWER
−6

GRADING NOTE Any equivalent form. x^2 terms: $x^2 \cdot 5 + 3x \cdot (-x) + (-2) \cdot 4x^2 = 5 - 3 - 8 = -6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q1/5

For which k does $(x + k)(x^2 - 3x + 1)$ have no x^2 term?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q2/5

515

QUESTION ID
S14-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. x^2 coefficient: $-3 + k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q2/5

True or false?

The constant term of a product of polynomials is the product of their constant terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q3/5

516

QUESTION ID
S14-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Only constants multiply to a constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q3/5

The leading coefficient of $(2x^3 - 1)(5x^2 + x)$ is

A. 7

B. 5

C. 2

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q4/5

517

QUESTION ID
S14-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 10

GRADING NOTE $2 \cdot 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q4**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q4/5

Find the coefficient of x in (x − 1)(x − 2)(x − 3).

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q5/5

518

QUESTION ID
S14-F1-Q5

TYPE
Short answer

CORRECT ANSWER
11

GRADING NOTE Any equivalent form. 1·2 + 1·3 + 2·3 = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q5**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 2

Expand powers of binomials with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 1)^4$ using Pascal's Triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q1/5

520

QUESTION ID
S14-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^4 + 4x^3 + 6x^2 + 4x + 1$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q1/5

Expand $(x - 2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q2/5

521

QUESTION ID
S14-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$x^4 - 8x^3 + 24x^2 - 32x + 16$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

The coefficients of $(x + y)^5$ are 1, 5, 10, 5, 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q3/5

522

QUESTION ID
S14-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Row 5 of Pascal's Triangle has six entries: 1, 5, 10, 10, 5, 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q3/5

The coefficient of x^2y^2 in $(x + y)^4$ is

A. 2

B. 6

C. 4

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q4/5

523

QUESTION ID
S14-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 6

GRADING NOTE C(4, 2) = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q4/5

Explain why the coefficients of $(x + y)^n$ are symmetric.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q5/5

524

QUESTION ID
S14-F2-Q5

TYPE
Short answer

CORRECT ANSWER
C(n, k) = C(n, n − k): choosing k of the n factors to give x is the same as choosing the n − k that give y.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q5/5

Find remainders with the Remainder Theorem

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Evaluate a polynomial function	**HSF-IF.A.2**
BACKWARD 2	Powers of negative numbers	**7.NS.A.2**
BACKWARD 3	Dividend = divisor × quotient + remainder	**4.NBT.B.6**
FORWARD 1	Find unknown coefficients from given remainders	**HSA-APR.B.2**
FORWARD 2	Prove the Remainder Theorem	**HSA-APR.B.2**

SET 15 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Use the Remainder Theorem to find the remainder when $p(x) = x^3 - 4x^2 + 2x - 5$ is divided by $x - 3$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q1/5

527

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form. $p(3) = 27 - 36 + 6 - 5 = -8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q1/5

Use the Remainder Theorem to find the remainder when $p(x) = 2x^4 + x^3 - 3x + 7$ is divided by $x + 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q2/5

528

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

37

GRADING NOTE Any equivalent form. Divisor $x - (-2)$: $p(-2) = 32 - 8 + 6 + 7 = 37$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q2/5

Use the Remainder Theorem to find the remainder when $p(x) = x^{100} + 3$ is divided by $x + 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q3/5

529

QUESTION ID

S15-M-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $p(-1) = (-1)^{100} + 3 = 1 + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-M-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q3/5

Use the Remainder Theorem to find the remainder when $p(x) = 4x^3 - 6x^2 + x - 1$ is divided by $x - 1/2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q4/5

530

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form (−1.5). p(1/2) = 1/2 − 3/2 + 1/2 − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q4/5

Use the Remainder Theorem to find the remainder when $p(x) = x^4 - 5x^2 + 4x$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q5/5

531

QUESTION ID

S15-M-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $p(2) = 16 - 20 + 8 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 1

Evaluate a polynomial function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = x^3 - 4x^2 + 2x - 5$. Find $p(1)$.

$p(1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q1/5

533

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q1/5

$p(x) = 2x^2 - x + 3$. Find $p(-2)$.

$p(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q2/5

534

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. 8 + 2 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q2/5

True or false?

If $p(x) = x^2 + 1$, then $p(-3) = -8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q3/5

535

QUESTION ID

S15-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-3)^2 + 1 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q3/5

If $p(x) = x^3 - x$, then $p(2) =$

A. 4

B. 6

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q4/5

536

QUESTION ID

S15-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE 8 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q4/5

$p(x) = 4x - 1$. Find $p(1/2)$.

$p(1/2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q5/5

537

QUESTION ID

S15-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 2

Powers of negative numbers • Grade 7 • The Number System • 7.NS.A.2

Evaluate $(-2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q1/5

539

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q1/5

Evaluate $(-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q2/5

540

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q2/5

$(-1)^{100}$ equals

A. -100

B. -1

C. 100

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q3/5

541

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE An even power of -1 is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q3/5

True or false?

$-2^4 = (-2)^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q4/5

542

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-2^4 = -16$ but $(-2)^4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $2(-2)^4 + (-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q5/5

543

QUESTION ID

S15-B2-Q5

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. 32 − 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 3

Dividend = divisor × quotient + remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Write 47 = 5 × q + r with 0 ≤ r < 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

545

QUESTION ID

S15-B3-Q1

TYPE

Short answer

CORRECT ANSWER

47 = 5 × 9 + 2

GRADING NOTE Form required: q = 9 and r = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

A division gives quotient 12 and remainder 3 with divisor 7. What was the dividend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

546

QUESTION ID

S15-B3-Q2

TYPE

Short answer

CORRECT ANSWER

87

GRADING NOTE Any equivalent form. 7 × 12 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

True or false?

If a division leaves remainder 0, the divisor is a factor of the dividend.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

547

QUESTION ID

S15-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Dividend = divisor × quotient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

$x^3 - 1 = (x - 1)(x^2 + x + 1) + 0$. The remainder on division by $x - 1$ is

A. 0

B. 1

C. $x^2 + x + 1$

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

548

QUESTION ID

S15-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0

GRADING NOTE Read from the identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

$p(x) = (x - 3)(x^2 - x - 1) - 8$. What is the remainder when $p(x)$ is divided by $x - 3$?

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

549

QUESTION ID

S15-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 1

Find unknown coefficients from given remainders • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

When $p(x) = x^3 + kx^2 - 4$ is divided by $x - 2$, the remainder is 12. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q1/5

551

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. $p(2) = 8 + 4k - 4 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = x^3 + ax + b$ leaves remainder 5 on division by $x - 1$ and remainder -1 on division by $x + 1$. Find a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q2/5

552

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 2, b = 2

GRADING NOTE Form required: both. $1 + a + b = 5$ and $-1 - a + b = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q2/5

True or false?

If p(x) leaves remainder 0 on division by x − 4, then p(−4) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q3/5

553

QUESTION ID

S15-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The remainder on division by x − 4 is p(4), so p(4) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q3/5

$p(x) = 2x^2 + kx + 1$ leaves remainder 7 on division by $x + 1$. Then $k =$

A. 4

B. -4

C. -6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q4/5

554

QUESTION ID

S15-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −4

GRADING NOTE $2 - k + 1 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q4/5

For which k is the remainder of $x^4 - kx + 6$ divided by $x + 2$ equal to 0?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q5/5

555

QUESTION ID

S15-F1-Q5

TYPE

Short answer

CORRECT ANSWER

−11

GRADING NOTE Any equivalent form. 16 + 2k + 6 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 2

Prove the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Prove that the remainder when a polynomial p(x) is divided by x − a is p(a).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q1/5

557

QUESTION ID

S15-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

p(a) = (a − a)q(a) + r = r.

GRADING NOTE REQUIRED WORK: Write p(x) = (x − a)q(x) + r and substitute x = a.

Any valid proof or derivation. Essential: the division identity with a constant remainder r (degree less than 1); substitution x = a; the factor (a − a) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q1/5

Why must the remainder on division by x − a be a constant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q2/5

558

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The remainder has a lower degree than the divisor x − a, which has degree 1, so it has degree 0 (a constant).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q2/5

True or false?

The Remainder Theorem also works for division by 2x − 1 with remainder p(1/2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q3/5

559

QUESTION ID

S15-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE p(x) = (2x − 1)q(x) + r gives p(1/2) = r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q3/5

Which step uses the fact that a − a = 0?

A. Graphing p(x)

B. Dividing p(x) by x

C. Factoring p(x)

D. Substituting x = a into p(x) = (x − a)q(x) + r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q4/5

560

QUESTION ID

S15-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Substituting x = a into p(x) = (x − a)q(x) + r

GRADING NOTE The product term vanishes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q4/5

Explain how the proof shows that p(a) = 0 exactly when x − a is a factor of p(x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q5/5

561

QUESTION ID

S15-F2-Q5

TYPE

Short answer

CORRECT ANSWER

p(a) = r, and r = 0 exactly when p(x) = (x − a)q(x), i.e. x − a is a factor.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q5/5

Use p(a) = 0 to find factors

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Use the Remainder Theorem	**HSA-APR.B.2**
BACKWARD 2	Factor the quadratic quotient	**HSA-SSE.A.2**
BACKWARD 3	Divide a polynomial by a linear factor	**HSA-APR.A.1**
FORWARD 1	Choose parameters so that a given binomial is a factor	**HSA-APR.B.2**
FORWARD 2	Finish factorizations over the complex numbers	**HSN-CN.C.8**

SET 16 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Show that $x - 2$ is a factor of $p(x) = x^3 - 3x^2 - 4x + 12$, then factor $p(x)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q1/5

564

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

$p(2) = 0$; $p(x) = (x - 2)(x + 2)(x - 3)$

GRADING NOTE Form required: the check and the complete factorization. $p(2) = 8 - 12 - 8 + 12 = 0$; quotient $x^2 - x - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q1/5

Show that x + 1 is a factor of $p(x) = x^3 + 2x^2 - 5x - 6$, then factor p(x) completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q2/5

565

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

p(−1) = 0; p(x) = (x + 1)(x + 3)(x − 2)

GRADING NOTE Form required: the check and the factorization. Quotient $x^2 + x - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-M-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q2/5

Show that $x + 3$ is a factor of $p(x) = 2x^3 + 5x^2 - 4x - 3$, then factor $p(x)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q3/5

566

QUESTION ID

S16-M-Q3

TYPE

Short answer

CORRECT ANSWER

$p(-3) = 0$; $p(x) = (x + 3)(2x + 1)(x - 1)$

GRADING NOTE Form required: the check and the factorization. $p(-3) = -54 + 45 + 12 - 3 = 0$; quotient $2x^2 - x - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q3/5

Is $x - 4$ a factor of $p(x) = x^3 - 5x^2 + 2x + 9$? Justify with the Remainder Theorem.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q4/5

567

QUESTION ID

S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

No: $p(4) = 1 \neq 0$

GRADING NOTE Form required: the decision and $p(4)$. $64 - 80 + 8 + 9 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-M-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q4/5

$p(x) = x^3 - 7x + 6$. Use $p(a) = 0$ to find all the zeros of p, and write $p(x)$ in factored form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q5/5

568

QUESTION ID

S16-M-Q5

TYPE

Short answer

CORRECT ANSWER

Zeros 1, 2, −3; $p(x) = (x - 1)(x - 2)(x + 3)$

GRADING NOTE Form required: all zeros and the factorization. $p(1) = 0$, then the quotient $x^2 + x - 6 = (x - 2)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 1

Use the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder when $x^3 - 3x^2 - 4x + 12$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q1/5

570

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q1/5

Find the remainder when $x^3 + 2x^2 - 5x - 6$ is divided by $x - 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q2/5

571

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q2/5

True or false?

Dividing by x + 3 means evaluating at x = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q3/5

572

QUESTION ID

S16-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x + 3 = x − (−3): evaluate at −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B1-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q3/5

The remainder of p(x) ÷ (x + 1) is

A. −p(1)

B. p(−1)

C. p(0)

D. p(1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q4/5

573

QUESTION ID

S16-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. p(−1)

GRADING NOTE x − (−1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $p(-3)$ for $p(x) = 2x^3 + 5x^2 - 4x - 3$.

$p(-3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q5/5

574

QUESTION ID
S16-B1-Q5

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 2

Factor the quadratic quotient • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q1/5

576

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 2)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $2x^2 - x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q2/5

577

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(2x + 1)(x − 1)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q2/5

$x^2 + x - 6$ equals

A. $(x - 3)(x + 2)$

B. $(x - 6)(x + 1)$

C. $(x + 3)(x - 2)$

D. $(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q3/5

578

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (x + 3)(x − 2)

GRADING NOTE Product −6, sum 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x^2 + 4$ factors over the integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q4/5

579

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q5/5

580

QUESTION ID

S16-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - 3)(x + 3)$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 3

Divide a polynomial by a linear factor • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find q(x) if $x^3 - 3x^2 - 4x + 12 = (x - 2)\,q(x)$.

q(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q1/5

582

QUESTION ID

S16-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 - x - 6$

GRADING NOTE Any equivalent form. Check by multiplying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q1/5

Multiply $(x + 1)(x^2 + x - 6)$ to check a division.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q2/5

583

QUESTION ID

S16-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + 2x^2 - 5x - 6$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q2/5

True or false?

If $(x - 1)(x^2 + x - 6) = x^3 - 7x + 6$, then $x^2 + x - 6$ is the quotient of $x^3 - 7x + 6 \div (x - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q3/5

584

QUESTION ID

S16-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of the quotient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^3 - 8 \div (x - 2)$ equals

A. $x^2 + 2x + 4$

B. $x^2 + 4$

C. $x^2 - 4$

D. $x^2 - 2x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q4/5

585

QUESTION ID

S16-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 2x + 4$

GRADING NOTE Difference of cubes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find q(x) if $2x^3 + 5x^2 - 4x - 3 = (x + 3)\,q(x)$.

q(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q5/5

586

QUESTION ID

S16-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2 - x - 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 1

Choose parameters so that a given binomial is a factor • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find k so that $x - 3$ is a factor of $x^3 + kx^2 - 3x + 9$.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q1/5

588

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form. $27 + 9k - 9 + 9 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q1/5

Find a and b so that $x - 1$ and $x + 2$ are both factors of $x^3 + ax^2 + bx - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q2/5

589

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 2, b = −1

GRADING NOTE Form required: both. $1 + a + b - 2 = 0$ and $-8 + 4a - 2b - 2 = 0$ give $a + b = 1$ and $2a - b = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q2/5

True or false?

If $x - 2$ and $x + 2$ are factors of $p(x)$, then $x^2 + 4$ is a factor of $p(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q3/5

590

QUESTION ID

S16-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x - 2)(x + 2) = x^2 - 4$, not $x^2 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q3/5

For which k is x + 1 a factor of $x^4 + kx + 3$?

A. 2

B. −4

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q4/5

591

QUESTION ID

S16-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE $1 - k + 3 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q4/5

Write a cubic with leading coefficient 1 that has $x - 1$, $x + 1$, and $x - 4$ as factors, in standard form.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q5/5

592

QUESTION ID

S16-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^3 - 4x^2 - x + 4$

GRADING NOTE Form required: standard form. $(x^2 - 1)(x - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 2

Finish factorizations over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

$x - 1$ is a factor of $x^3 - 1$. Factor $x^3 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q1/5

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1/2 - (\sqrt{3}/2)i)(x + 1/2 + (\sqrt{3}/2)i)$

GRADING NOTE Any equivalent form. The quotient $x^2 + x + 1$ has zeros $-1/2 \pm (\sqrt{3}/2)i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$x + 2$ is a factor of $x^3 + 2x^2 + 9x + 18$. Factor it completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q2/5

595

QUESTION ID
S16-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$(x + 2)(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q2/5

True or false?

Every factor found with the factor theorem gives a zero of the polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q3/5

596

QUESTION ID

S16-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x - a$ is a factor exactly when $p(a) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q3/5

$x - 2$ is a factor of $x^3 - 2x^2 + x - 2$. The other zeros are

A. i and −i

B. −2

C. 1 and −1

D. 2i and −2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q4/5

597

QUESTION ID

S16-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. i and −i

GRADING NOTE Quotient $x^2 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q4/5

$x - 3$ is a factor of $x^3 - 3x^2 + 4x - 12$. List all complex zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q5/5

598

QUESTION ID
S16-F2-Q5

TYPE
Short answer

CORRECT ANSWER
3, 2i, −2i

GRADING NOTE Form required: all three. Quotient $x^2 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q5/5

Identify zeros and multiplicities from a factorization

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Solve by the zero-product property	**HSA-REI.B.4.b**
BACKWARD 2	Factor to find hidden repeated factors	**HSA-SSE.A.2**
BACKWARD 3	An exponent counts repeated factors	**6.EE.A.1**
FORWARD 1	Build a polynomial from zeros and one more condition	**HSA-APR.B.3**
FORWARD 2	Decide what the zeros determine and what remains free	**HSA-APR.B.3**

SET 17 • HSA-APR.B.3

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $p(x) = (x - 4)^2(x + 1)(2x - 3)$ and give the multiplicity of each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q1/5

601

QUESTION ID
S17-M-Q1

TYPE
Short answer

CORRECT ANSWER
4 (multiplicity 2), −1 (multiplicity 1), 3/2 (multiplicity 1)

GRADING NOTE Form required: every zero with its multiplicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q1/5

Factor $p(x) = x^3 - 6x^2 + 9x$, then list its zeros with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q2/5

602

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

$p(x) = x(x - 3)^2$; 0 (multiplicity 1), 3 (multiplicity 2)

GRADING NOTE Form required: factorization and zeros with multiplicities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q2/5

List the zeros of $p(x) = (x^2 - 9)(x^2 + 2x + 1)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q3/5

603

QUESTION ID

S17-M-Q3

TYPE

Short answer

CORRECT ANSWER

3 (1), −3 (1), −1 (2)

GRADING NOTE Form required: every zero with its multiplicity. $(x - 3)(x + 3)(x + 1)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-M-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q3/5

List the zeros of $p(x) = -2x(x + 5)^3$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q4/5

604

QUESTION ID

S17-M-Q4

TYPE

Short answer

CORRECT ANSWER

0 (1), −5 (3)

GRADING NOTE Form required: every zero with its multiplicity. The constant −2 is not a zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q4/5

Factor $p(x) = x^4 - 8x^2 + 16$, then list its zeros with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q5/5

605

QUESTION ID

S17-M-Q5

TYPE

Short answer

CORRECT ANSWER

$p(x) = (x - 2)^2(x + 2)^2$; 2 (2), −2 (2)

GRADING NOTE Form required: factorization and zeros with multiplicities. $(x^2 - 4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 1

Solve by the zero-product property • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 4)(x + 1) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q1/5

607

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 or −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve x(2x − 3) = 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q2/5

608

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0 or 3/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q2/5

True or false?

If $(x - 2)^2 = 0$, then $x = 2$ or $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q3/5

609

QUESTION ID

S17-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solutions of $(x + 5)(x - 5) = 0$ are

A. 25

B. −5 and 5

C. 0 and 5

D. 5 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q4/5

610

QUESTION ID

S17-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −5 and 5

GRADING NOTE Each factor equal to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-2x(x + 5) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q5/5

611

QUESTION ID

S17-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0 or −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 2

Factor to find hidden repeated factors • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 6x + 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q1/5

613

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q1/5

Factor $x^2 + 2x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q2/5

614

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^4 - 8x^2 + 16$ equals

A. $(x^2 + 4)^2$

B. $(x^2 - 4)^2$

C. $(x^2 - 8)^2$

D. $(x - 4)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q3/5

615

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x^2 - 4)^2$

GRADING NOTE $u^2 - 8u + 16$ with $u = x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x^3 - 6x^2 + 9x = x(x + 3)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q4/5

616

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x(x + 3)^2 = x^3 + 6x^2 + 9x$; the correct form is $x(x - 3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q5/5

617

QUESTION ID

S17-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q5**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 3

An exponent counts repeated factors • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $(x - 4)^2$ as a product without an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q1/5

619

QUESTION ID

S17-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 4)(x − 4)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many factors of $(x + 5)$ are in $(x + 5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q2/5

620

QUESTION ID

S17-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q2/5

True or false?

$(x - 2)^2(x - 2) = (x - 2)^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q3/5

621

QUESTION ID

S17-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q3/5

In $7(x - 1)^4$, the factor $x - 1$ appears

A. 28 times

B. 4 times

C. 1 time

D. 7 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q4/5

622

QUESTION ID

S17-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4 times

GRADING NOTE Exponent 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (x + 1)(x + 1)(x − 3) with an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q5/5

623

QUESTION ID

S17-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2(x - 3)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 1

Build a polynomial from zeros and one more condition • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Write the cubic polynomial with zeros −1, 2, and 4 (each multiplicity 1) whose graph passes through (0, 16).

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q1/5

625

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2(x + 1)(x − 2)(x − 4)

GRADING NOTE Any equivalent form. a(1)(−2)(−4) = 8a = 16 gives a = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q1/5

Write a polynomial of degree 3 with a zero of multiplicity 2 at x = 3, a zero at x = −2, and leading coefficient −1.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q2/5

626

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$-(x-3)^2(x+2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q2/5

True or false?

Zeros alone determine a polynomial completely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q3/5

627

QUESTION ID

S17-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any nonzero constant multiple has the same zeros; a further condition fixes the scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q3/5

A quartic has zeros 1 and −1, each of multiplicity 2, and p(0) = 3. Then p(x) =

A. $-3(x-1)^2(x+1)^2$

B. $3(x^2-1)$

C. $(x-1)^2(x+1)^2$

D. $3(x-1)^2(x+1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q4/5

628

QUESTION ID

S17-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $3(x-1)^2(x+1)^2$

GRADING NOTE $a \cdot 1 \cdot 1 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q4/5

A polynomial has zeros 0 (multiplicity 2) and 5 (multiplicity 1), degree 3, and p(1) = −8. Find p(x).

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q5/5

629

QUESTION ID

S17-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2(x - 5)$

GRADING NOTE Any equivalent form. a · 1 · (−4) = −8 gives a = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 2

Decide what the zeros determine and what remains free • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

All polynomials with zeros 1 and 3 only (each multiplicity 1) and degree 2 have what form?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q1/5

631

QUESTION ID
S17-F2-Q1

TYPE
Short answer

CORRECT ANSWER
a(x − 1)(x − 3) with any real a ≠ 0

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q1/5

If the degree is not fixed, give two different polynomials whose only real zeros are 1 and 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q2/5

632

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

For example $(x - 1)(x - 3)$ and $(x - 1)^2(x - 3)(x^2 + 1)$

GRADING NOTE Any two valid examples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q2/5

True or false?

Two cubic polynomials with the same three zeros (each multiplicity 1) must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q3/5

633

QUESTION ID

S17-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They can differ by a constant factor, for example $(x-1)(x-2)(x-3)$ and $2(x-1)(x-2)(x-3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q3/5

Which extra information fixes a cubic with zeros −2, 0, 2?

A. That it has degree 3

B. That 0 is a zero

C. Its end behavior only

D. Its value at x = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q4/5

634

QUESTION ID

S17-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Its value at x = 1

GRADING NOTE One point fixes the scale a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q4/5

Explain why knowing only the real zeros of a degree-4 polynomial may not determine it even up to scale.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q5/5

635

QUESTION ID

S17-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It may have non-real zeros (an irreducible quadratic factor such as $x^2 + c$) that the real zeros do not show, so different quartics share the same real zeros.

GRADING NOTE Any equivalent explanation with an example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q5/5

Construct a rough graph of a polynomial from its zeros

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Read zeros and multiplicities	**HSA-APR.B.3**
BACKWARD 2	End behavior from the leading term	**HSF-IF.B.4**
BACKWARD 3	Find the y-intercept by evaluating at 0	**HSF-IF.A.2**
BACKWARD 4	Sign of a power of a negative or positive number	**7.NS.A.2**
FORWARD 1	Sketch graphs from standard form by factoring first	**HSF-IF.C.7.c**
FORWARD 2	Write a polynomial from its rough graph	**HSA-APR.B.3**

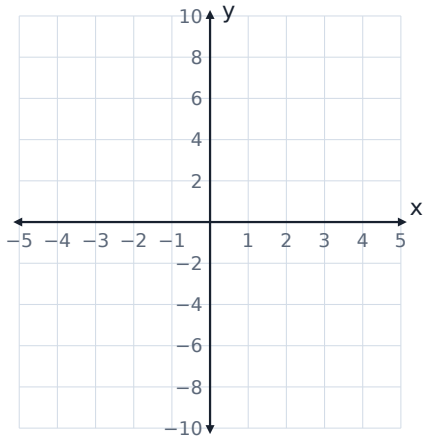
SET 18 • HSA-APR.B.3

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Sketch a rough graph on the grid. Show every zero (crossing or touching the x-axis), the y-intercept, and the end behavior.

$y = (x + 2)(x - 1)(x - 3)$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q1/5

638

QUESTION ID

S18-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Crosses at x = −2, 1, 3; y-intercept 6; falls to the left and rises to the right (degree 3, positive leading coefficient).

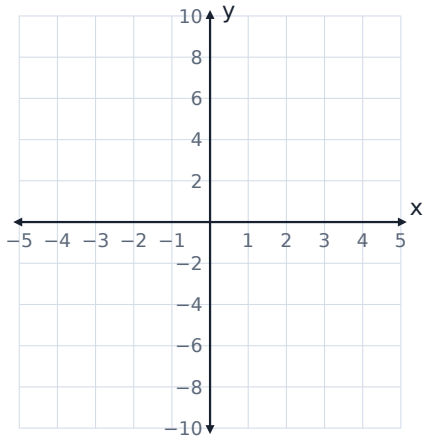
GRADING NOTE Any rough graph with all listed features: the three crossings, (0, 6), and both ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q1/5

Sketch a rough graph on the grid. Show every zero (crossing or touching the x-axis), the y-intercept, and the end behavior.

$y = -(x + 1)^2(x - 2)$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q2/5

639

QUESTION ID

S18-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Touches at x = −1 (even multiplicity) and crosses at x = 2; y-intercept 2; rises to the left and falls to the right (degree 3, negative leading coefficient).

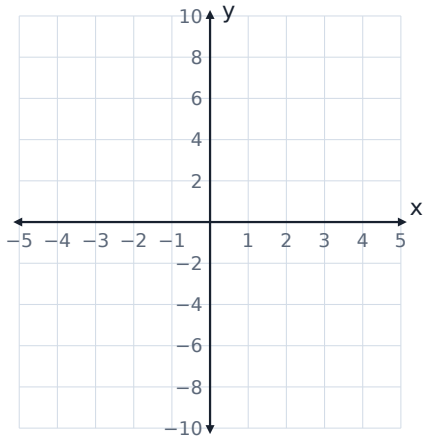
GRADING NOTE Any rough graph with all listed features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q2/5

Sketch a rough graph on the grid. Show every zero (crossing or touching the x-axis), the y-intercept, and the end behavior.

$y = x(x - 3)^2$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q3/5

640

QUESTION ID

S18-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Crosses at x = 0 and touches at x = 3; y-intercept 0; falls to the left and rises to the right.

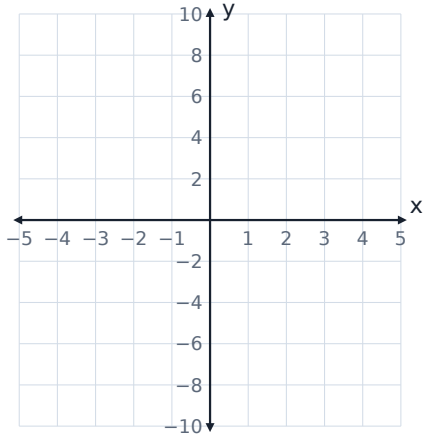
GRADING NOTE Any rough graph with all listed features. Between 0 and 3 the graph is above the axis (for example y(1) = 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q3/5

Sketch a rough graph on the grid. Show every zero (crossing or touching the x-axis), the y-intercept, and the end behavior.

$y = (x+2)(x+1)(x-1)(x-3)$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q4/5

641

QUESTION ID

S18-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Crosses at x = −2, −1, 1, 3; y-intercept 6; rises at both ends (degree 4, positive leading coefficient).

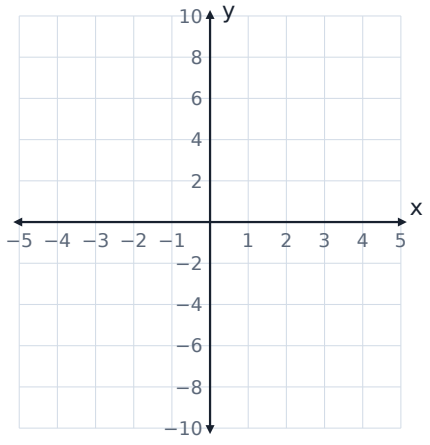
GRADING NOTE Any rough graph with all listed features. Signs: positive for $x < -2$, negative on $(-2, -1)$, positive on $(-1, 1)$, negative on $(1, 3)$, positive for $x > 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q4/5

Sketch a rough graph on the grid. Show every zero (crossing or touching the x-axis), the y-intercept, and the end behavior.

$y = -(x - 1)^3(x + 2)$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q5/5

642

QUESTION ID

S18-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Crosses at x = −2 and crosses at x = 1 with a flattened (inflection) shape (multiplicity 3); y-intercept 2; falls at both ends (degree 4, negative leading coefficient).

GRADING NOTE Any rough graph with all listed features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 1

Read zeros and multiplicities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q1/5

644

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$-2, 1, 3$

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q1/5

List the zeros of $-(x + 1)^2(x - 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q2/5

645

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−1 (2), 2 (1)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q2/5

True or false?

$x(x - 3)^2$ has a zero of multiplicity 2 at $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q3/5

646

QUESTION ID

S18-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 0$ has multiplicity 1; $x = 3$ has multiplicity 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q3/5

The zero $x = 1$ of $-(x - 1)^3(x + 2)$ has multiplicity

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q4/5

647

QUESTION ID

S18-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE Exponent 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the zeros of $(x + 2)(x + 1)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q5/5

QUESTION ID

S18-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2, −1, 1, 3

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 2

End behavior from the leading term • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

As x grows large and positive, what happens to x^3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q1/5

650

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

It grows large and positive.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

As x grows large and negative, what happens to $-x^4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q2/5

651

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

It becomes large and negative.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q2/5

The leading term of $(x + 2)(x - 1)(x - 3)$ is

A. $-x^3$

B. -6

C. $3x$

D. x^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q3/5

652

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x^3

GRADING NOTE $x \cdot x \cdot x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q3/5

True or false?

A polynomial of even degree with a positive leading coefficient falls at both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q4/5

653

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Like x^2, it rises at both ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the leading term of $-(x-1)^3(x+2)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q5/5

654

QUESTION ID

S18-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$-x^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 3

Find the y-intercept by evaluating at 0 • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find y when x = 0 for y = (x + 2)(x − 1)(x − 3).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q1/5

656

QUESTION ID

S18-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 2 · (−1) · (−3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q1/5

Find y when x = 0 for $y = -(x + 1)^2(x - 2)$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q2/5

657

QUESTION ID

S18-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. −(1)(−2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q2/5

True or false?

The y-intercept of $x(x - 3)^2$ is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q3/5

658

QUESTION ID

S18-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 0 (a factor x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q3/5

The y-intercept of $-(x-1)^3(x+2)$ is

A. 0

B. -2

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q4/5

659

QUESTION ID

S18-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $-(-1)(2) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q4/5

Find y when x = 1 for $y = x(x - 3)^2$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q5/5

660

QUESTION ID

S18-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 4

Sign of a power of a negative or positive number • Grade 7 • The Number System • 7.NS.A.2

Is $(-0.5)^2$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q1/5

662

QUESTION ID

S18-B4-Q1

TYPE

Short answer

CORRECT ANSWER

Positive

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q1/5

Is $(-0.5)^3$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q2/5

663

QUESTION ID

S18-B4-Q2

TYPE

Short answer

CORRECT ANSWER

Negative

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q2/5

True or false?

An even power of a negative number is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q3/5

664

QUESTION ID

S18-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pairs of negative factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q3/5

Near $x = 3$, $(x - 3)^2$ is

A. negative on both sides

B. positive on both sides

C. zero everywhere

D. positive on the left and negative on the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q4/5

665

QUESTION ID

S18-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. positive on both sides

GRADING NOTE A square does not change sign at $x = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q4/5

Does $(x - 1)^3$ change sign at $x = 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q5/5

666

QUESTION ID

S18-B4-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: negative for $x < 1$ and positive for $x > 1$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q5/5

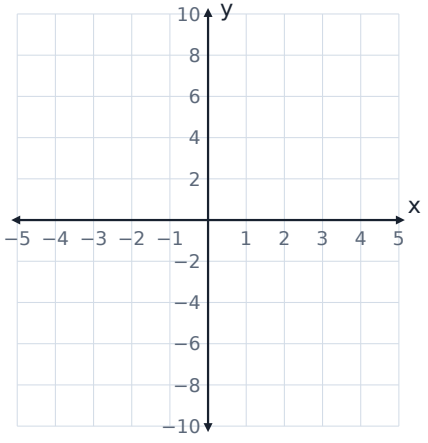
SET 18 • HSA-APR.B.3

FORWARD 1

Sketch graphs from standard form by factoring first • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Factor, then sketch a rough graph showing every zero, the y-intercept, and end behavior.

$y = x^3 - x^2 - 6x$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q1/5

668

QUESTION ID

S18-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

x(x − 3)(x + 2): crosses at −2, 0, 3; y-intercept 0; falls left and rises right.

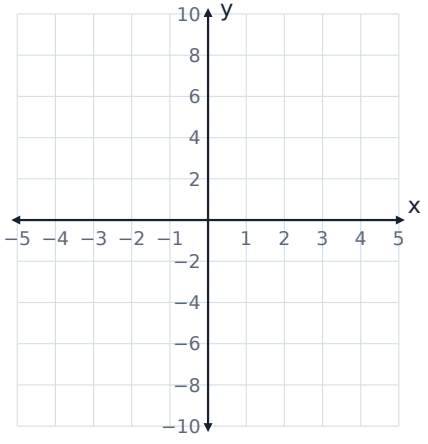
GRADING NOTE Any rough graph with all features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q1/5

Factor, then sketch a rough graph showing every zero, the y-intercept, and end behavior.

$y = x^4 - 5x^2 + 4$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q2/5

669

QUESTION ID

S18-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

(x − 1)(x + 1)(x − 2)(x + 2): crosses at ±1, ±2; y-intercept 4; rises at both ends.

GRADING NOTE Any rough graph with all features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q2/5

True or false?

$y = x^3 + 4x$ crosses the x-axis at $x = -2$, 0, and 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q3/5

670

QUESTION ID

S18-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x(x^2 + 4) = 0$ only at $x = 0$; $x^2 + 4$ has no real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q3/5

$y = -x^3 + 9x$ crosses the x-axis at

A. 0 and 9

B. 3 only

C. −3, 0, 3

D. −9, 0, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q4/5

671

QUESTION ID

S18-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −3, 0, 3

GRADING NOTE $-x(x - 3)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q4/5

$y = x^3 - 3x^2 + 4 = (x + 1)(x - 2)^2$. Describe its behavior at each zero.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q5/5

672

QUESTION ID

S18-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Crosses at x = −1; touches (turns) at x = 2.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

FORWARD 2

Write a polynomial from its rough graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A cubic's graph crosses the x-axis at −3 and touches it at 2, and passes through (0, 12). Write the polynomial.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q1/5

674

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(x - 2)^2$

GRADING NOTE Any equivalent form. a(3)(4) = 12 gives a = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q1/5

A quartic rises at both ends, touches the x-axis at −1 and at 3, and passes through (0, 18). Write the polynomial.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q2/5

675

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$2(x + 1)^2(x - 3)^2$

GRADING NOTE Any equivalent form. a(1)(9) = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q2/5

True or false?

A graph that touches the x-axis at x = a without crossing has a zero of even multiplicity at a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q3/5

676

QUESTION ID

S18-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The factor does not change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q3/5

A cubic falls to the right and crosses the x-axis at 0, 1, and 4. Its leading coefficient is

A. zero

B. not determined by the end behavior

C. positive

D. negative

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q4/5

677

QUESTION ID

S18-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. negative

GRADING NOTE A cubic that falls to the right has a negative leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q4/5

A polynomial graph crosses at −2, touches at 1, falls to the left and rises to the right, and has y-intercept 4. Give a polynomial of least degree.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q5/5

678

QUESTION ID

S18-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2(x + 2)(x - 1)^2$

GRADING NOTE Any equivalent form. Degree 3 with a > 0; a(2)(1) = 4 gives a = 2, which agrees with the end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q5/5

Prove polynomial identities

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Expand the square of a binomial	**HSA-APR.A.1**
BACKWARD 2	Recognize a difference of squares	**HSA-SSE.A.2**
BACKWARD 3	What makes an identity	**6.EE.A.4**
FORWARD 1	Prove or disprove proposed identities	**HSA-APR.C.4**
FORWARD 2	Prove identities with the Binomial Theorem	**HSA-APR.C.5**

SET 19 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Prove the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q1/5

681

QUESTION ID

S19-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Left: $x^4 + 2x^2y^2 + y^4$. Right: $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The two sides are equal.

GRADING NOTE

REQUIRED WORK: Expand both sides (or one side) and show they are equal.

Any valid proof or derivation. Essential: correct expansion of both squares and the combination $-2x^2y^2 + 4x^2y^2 = 2x^2y^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q1/5

Prove the identity $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q2/5

682

QUESTION ID

S19-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$(a + b)(a^2 - ab + b^2) = a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3 = a^3 + b^3$.

GRADING NOTE REQUIRED WORK: Multiply out the right side.

Any valid proof or derivation. Essential: all six products and the cancellation of $-a^2b + a^2b$ and $ab^2 - ab^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove the identity $(a + b)^2 - (a - b)^2 = 4ab$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q3/5

683

QUESTION ID

S19-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab.$
(Or:
$[(a + b) - (a - b)][(a + b) + (a - b)]$
$= (2b)(2a) = 4ab.$)

GRADING NOTE REQUIRED WORK: Expand both squares, or factor as a difference of squares.

Any valid proof or derivation. Essential: subtracting the whole second square (the sign of every term changes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q3/5

Prove the identity $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q4/5

QUESTION ID

S19-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$(x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4.$

GRADING NOTE REQUIRED WORK: Multiply the right side, or write it as a difference of squares.

Any valid proof or derivation. Essential: grouping $(x^2 + 2 + 2x)(x^2 + 2 - 2x)$ as a difference of squares, or a complete expansion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-M-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q4/5

Prove the identity $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q5/5

685

QUESTION ID

S19-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$(n + 1)^3 = n^3 + 3n^2 + 3n + 1$, so $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$.

GRADING NOTE REQUIRED WORK: Expand $(n + 1)^3$ and simplify.

Any valid proof or derivation. Essential: the correct expansion of the cube.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 1

Expand the square of a binomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q1/5

687

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$a^2 + 2ab + b^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x^2 - y^2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q2/5

688

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^4 - 2x^2y^2 + y^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q2/5

True or false?

$(a - b)^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q3/5

689

QUESTION ID

S19-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(a - b)^2 = a^2 - 2ab + b^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(2xy)^2$ equals

A. $2xy^2$

B. $4xy$

C. $2x^2y^2$

D. $4x^2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q4/5

690

QUESTION ID

S19-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $4x^2y^2$

GRADING NOTE Square each factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q4/5

Expand $(n + 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q5/5

691

QUESTION ID

S19-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$n^3 + 3n^2 + 3n + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 2

Recognize a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $a^2 - b^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q1/5

693

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(a − b)(a + b)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $(x^2 + 2)^2 - (2x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q2/5

694

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 2 - 2x)(x^2 + 2 + 2x)$

GRADING NOTE Any equivalent factored form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q2/5

True or false?

$u^2 - v^2 = (u - v)(u + v)$ for any expressions u and v.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q3/5

695

QUESTION ID

S19-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Structure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q3/5

Which is a difference of squares?

A. $x^3 - 1$

B. $x^2 - 3x$

C. $9x^2 - 16$

D. $9x^2 + 16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q4/5

696

QUESTION ID

S19-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $9x^2 - 16$

GRADING NOTE $(3x)^2 - 4^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q4/5

Factor $25 - x^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q5/5

697

QUESTION ID

S19-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(5 - x^2)(5 + x^2)$

GRADING NOTE Any equivalent factored form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 3

What makes an identity • Grade 6 • Expressions & Equations • 6.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Test $(a + b)^2 = a^2 + b^2$ with $a = 1$ and $b = 2$. Is it an identity?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q1/5

699

QUESTION ID

S19-B3-Q1

TYPE

Short answer

CORRECT ANSWER

No: 9 ≠ 5

GRADING NOTE Form required: the test and the decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain the difference between an equation and an identity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q2/5

700

QUESTION ID

S19-B3-Q2

TYPE

Short answer

CORRECT ANSWER

An identity is true for every value of the variables; an equation may be true only for some values.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q2/5

True or false?

Checking an identity with three numbers proves it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q3/5

701

QUESTION ID

S19-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Examples cannot prove a statement about all values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q3/5

$2(x + 3) = 2x + 6$ is

A. an identity

B. true only for $x = 3$

C. true only for $x = 0$

D. never true

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q4/5

702

QUESTION ID

S19-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. an identity

GRADING NOTE Distributive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q4/5

Give values that show $a^3 + b^3 = (a + b)^3$ is not an identity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q5/5

703

QUESTION ID

S19-B3-Q5

TYPE

Short answer

CORRECT ANSWER

For example

$a = b = 1$: $2 \neq 8$

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 1

Prove or disprove proposed identities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove: $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q1/5

705

QUESTION ID

S19-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

True:
$3ab(a + b) = 3a^2b + 3ab^2$, and
$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.

GRADING NOTE REQUIRED WORK: Expand both sides, or give a counterexample.

Any valid proof or derivation. Essential: a correct decision with a full expansion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q1/5

Prove or disprove: $a^4 - b^4 = (a - b)^4$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q2/5

706

QUESTION ID

S19-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

False: a = 2, b = 1 gives 15 ≠ 1.

GRADING NOTE REQUIRED WORK: Give a counterexample or a proof.

Any valid proof or derivation. Essential: a numerical counterexample (a factorization $(a - b)(a + b)(a^2 + b^2)$ is also acceptable).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q2/5

True or false?

$(x - 1)(x^2 + x + 1) = x^3 + 1$ is an identity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q3/5

707

QUESTION ID

S19-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The product is $x^3 - 1$ (difference of cubes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q3/5

Which is an identity?

A. $(x + 2)^2 = x^2 + 4$

B. $(x + 1)^3 = x^3 + 1$

C. $x^2 - 4 = (x - 4)^2$

D. $(x + 2)^2 - (x - 2)^2 = 8x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q4/5

708

QUESTION ID

S19-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $(x + 2)^2 - (x - 2)^2 = 8x$

GRADING NOTE 4ab with a = x, b = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find c so that $(x + c)^2 - (x - c)^2 = 12x$ is an identity.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q5/5

709

QUESTION ID

S19-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 4cx = 12x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 2

Prove identities with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Use the Binomial Theorem with $x = y = 1$ to prove that the entries in row n of Pascal's Triangle add to 2^n.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q1/5

711

QUESTION ID

S19-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$(1 + 1)^n = \Sigma\ C(n, k)$
$1^k\ 1^{n-k} = \Sigma\ C(n, k)$,
so the row sum is 2^n.

GRADING NOTE REQUIRED WORK: Substitute into $(x + y)^n$.

Any valid proof or derivation. Essential: the substitution and the sum of the coefficients.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q1/5

Use the Binomial Theorem with $x = 1$ and $y = -1$ to find $C(5,0) - C(5,1) + C(5,2) - C(5,3) + C(5,4) - C(5,5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q2/5

712

QUESTION ID
S19-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE Any equivalent form. $(1 - 1)^5 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q2/5

True or false?

$(1 + 1)^4 = 1 + 4 + 6 + 4 + 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q3/5

713

QUESTION ID
S19-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q3/5

The sum of the coefficients of $(2x + 1)^3$ is

A. 9

B. 8

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q4/5

714

QUESTION ID
S19-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 27

GRADING NOTE Set x = 1: 3^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $(1 + 0.1)^3 = 1 + 3(0.1) + 3(0.01) + 0.001$ to find 1.1^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q5/5

715

QUESTION ID
S19-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1.331

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q5/5

Use polynomial identities to describe numerical relationships

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Substitute values into an identity	**6.EE.A.2**
BACKWARD 2	Check a Pythagorean triple	**8.G.B.6**
BACKWARD 3	Choose a and b to fit a product	**HSA-APR.C.4**
FORWARD 1	Generate a triple with a required side	**HSA-APR.C.4**
FORWARD 2	Connect the triple identity to complex numbers	**HSN-CN.A.2**

SET 20 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ generates Pythagorean triples. Use $x = 3$ and $y = 2$ to find a triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q1/5

718

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

5, 12, 13

GRADING NOTE Any order. $x^2 - y^2 = 5$, $2xy = 12$, $x^2 + y^2 = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q1/5

Use the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ with $x = 5$ and $y = 2$ to find a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q2/5

719

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

20, 21, 29

GRADING NOTE Any order. $25 - 4 = 21$, $2 \cdot 5 \cdot 2 = 20$, $25 + 4 = 29$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q2/5

Use the identity $(a + b)(a - b) = a^2 - b^2$ to compute 47×53 mentally. Show which a and b you use.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q3/5

720

QUESTION ID

S20-M-Q3

TYPE

Short answer

CORRECT ANSWER

a = 50, b = 3:

2500 − 9 = 2491

GRADING NOTE Form required: the choice of a and b and the product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q3/5

Use the identity $(n + 1)^2 - n^2 = 2n + 1$ to write 41 as a difference of two consecutive squares.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q4/5

721

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

$41 = 21^2 - 20^2$

GRADING NOTE Form required: the two squares. $2n + 1 = 41$ gives $n = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q4/5

Use the identity $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$ to compute $13^3 - 12^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q5/5

722

QUESTION ID
S20-M-Q5

TYPE
Short answer

CORRECT ANSWER
469

GRADING NOTE Any equivalent form. 3·144 + 36 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 1

Substitute values into an identity • Grade 6 • Expressions & Equations • 6.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $x^2 - y^2$ for $x = 3$ and $y = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q1/5

724

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2xy for x = 5 and y = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q2/5

725

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q2/5

True or false?

For $n = 12$, $3n^2 + 3n + 1 = 433$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q3/5

726

QUESTION ID

S20-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $432 + 36 + 1 = 469$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $a = 50$ and $b = 3$, $a^2 - b^2$ is

A. 2491

B. 2500

C. 2494

D. 2509

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q4/5

727

QUESTION ID

S20-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2491

GRADING NOTE 2500 − 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2n + 1 for n = 20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q5/5

728

QUESTION ID

S20-B1-Q5

TYPE

Short answer

CORRECT ANSWER

41

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 2

Check a Pythagorean triple • Grade 8 • Geometry • 8.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Do 5, 12, 13 satisfy $a^2 + b^2 = c^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q1/5

730

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: 25 + 144 = 169

GRADING NOTE Form required: the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Do 20, 21, 29 satisfy $a^2 + b^2 = c^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q2/5

731

QUESTION ID

S20-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes:

400 + 441 = 841

GRADING NOTE Form required: the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q2/5

True or false?

6, 8, 11 is a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q3/5

732

QUESTION ID

S20-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $36 + 64 = 100 \neq 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q3/5

Which is a Pythagorean triple?

A. 8, 15, 17

B. 6, 7, 9

C. 2, 3, 4

D. 4, 5, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q4/5

733

QUESTION ID

S20-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8, 15, 17

GRADING NOTE 64 + 225 = 289.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q4/5

Explain why a triangle with sides 5, 12, 13 is a right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q5/5

734

QUESTION ID

S20-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$5^2 + 12^2 = 13^2$, so by the converse of the Pythagorean Theorem it is a right triangle.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 3

Choose a and b to fit a product • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

ANSWER KEY Reduced copy of the question page, with its answer

Write 47 and 53 as 50 − b and 50 + b. What is b?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q1/5

736

QUESTION ID

S20-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q1/5

Write 98 × 102 as (a − b)(a + b).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q2/5

737

QUESTION ID

S20-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(100 − 2)(100 + 2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$29 \times 31 = 30^2 + 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q3/5

738

QUESTION ID

S20-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(30 - 1)(30 + 1) = 30^2 - 1 = 899.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q3/5

64 × 56 equals

A. $(60 - 4)^2$

B. $64^2 - 56^2$

C. $60^2 - 4^2$

D. $60^2 + 4^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q4/5

739

QUESTION ID

S20-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $60^2 - 4^2$

GRADING NOTE a = 60, b = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 98 × 102.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q5/5

740

QUESTION ID

S20-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9996

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 1

Generate a triple with a required side • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Use the triple identity to find x and y (whole numbers, x > y) that give a triple with hypotenuse 25 and legs 7 and 24.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q1/5

742

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 4, y = 3

GRADING NOTE Any equivalent form. $x^2 + y^2 = 25$, $x^2 - y^2 = 7$, $2xy = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find x and y (whole numbers, x > y) for which the identity gives a triple with a leg 2xy = 40 and hypotenuse 41.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q2/5

743

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 5, y = 4

GRADING NOTE Any equivalent form. $x^2 - y^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q2/5

True or false?

Every choice of whole numbers $x > y > 0$ in the identity gives a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q3/5

744

QUESTION ID

S20-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All three expressions are positive whole numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q3/5

Which choice gives the triple 8, 15, 17?

A. x = 4, y = 1

B. x = 5, y = 3

C. x = 4, y = 2

D. x = 3, y = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q4/5

745

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x = 4, y = 1

GRADING NOTE 16 − 1 = 15, 8, 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q4/5

Explain why $x = 2y$ gives a triple with legs in the ratio $3 : 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q5/5

746

QUESTION ID

S20-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^2 - y^2 = 3y^2$ and $2xy = 4y^2$, so the legs are $3y^2$ and $4y^2$ (ratio 3 : 4).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 2

Connect the triple identity to complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(3 + 2i)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q1/5

748

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5 + 12i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(x + yi)^2$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q2/5

749

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x^2 - y^2) + 2xyi$

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q2/5

True or false?

The real and imaginary parts of $(x + yi)^2$ are the legs of the Pythagorean triple from the identity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q3/5

750

QUESTION ID
S20-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $x^2 - y^2$ and $2xy$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q3/5

$(5 + 2i)^2$ equals

A. 25 + 4i

B. 21 + 20i

C. 21 + 10i

D. 29 + 20i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q4/5

751

QUESTION ID

S20-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 21 + 20i

GRADING NOTE 25 − 4 + 20i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q4/5

Use $(4 + i)^2$ to find a Pythagorean triple.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q5/5

752

QUESTION ID

S20-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15, 8, 17

GRADING NOTE Any order. $(4 + i)^2 = 15 + 8i$; $4^2 + 1^2 = 17$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q5/5

Expand with the Binomial Theorem

MAIN	Algebra II main question	**HSA-APR.C.5**
BACKWARD 1	Multiply binomials repeatedly	**HSA-APR.A.1**
BACKWARD 2	Build Pascal's Triangle by a rule	**4.OA.C.5**
BACKWARD 3	Powers of products and of negatives	**8.EE.A.1**
FORWARD 1	Find a specific term or coefficient with a condition	**HSA-APR.C.5**
FORWARD 2	Use binomial coefficients to compute probabilities	**HSS-CP.B.9**

SET 21 • HSA-APR.C.5

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Use the Binomial Theorem to expand $(x + 2)^5$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q1/5

755

QUESTION ID
S21-M-Q1

TYPE
Short answer

CORRECT ANSWER
$x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$

GRADING NOTE Form required: standard form. Coefficients $C(5, k)\cdot 2^k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q1**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q1/5

Use the Binomial Theorem to expand $(2a - b)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q2/5

756

QUESTION ID
S21-M-Q2

TYPE
Short answer

CORRECT ANSWER
$16a^4 - 32a^3b + 24a^2b^2 - 8ab^3 + b^4$

GRADING NOTE Any order of terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q2**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q2/5

Use the Binomial Theorem to find the coefficient of x^3 in $(x + 3)^7$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q3/5

757

QUESTION ID
S21-M-Q3

TYPE
Short answer

CORRECT ANSWER
2835

GRADING NOTE Any equivalent form. $C(7, 3)\cdot 3^4 = 35\cdot 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q3/5

Use the Binomial Theorem to find the term containing x^3y^3 in $(x - 2y)^6$.

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q4/5

758

QUESTION ID
S21-M-Q4

TYPE
Short answer

CORRECT ANSWER
$-160x^3y^3$

GRADING NOTE Any equivalent form. $C(6, 3)(-2)^3 = 20\cdot(-8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q4**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q4/5

Use the first three terms of the binomial expansion of $(1 + 0.02)^5$ to estimate 1.02^5 to four decimal places.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q5/5

759

QUESTION ID
S21-M-Q5

TYPE
Short answer

CORRECT ANSWER
1.1040

GRADING NOTE Any value 1.1040 or 1.1041 is accepted (1 + 0.1 + 0.004 = 1.104; the exact value is 1.10408...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 1

Multiply binomials repeatedly • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q1/5

761

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 4x + 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q2/5

762

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + 6x^2 + 12x + 8$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q2/5

True or false?

$(x + y)^3 = x^3 + y^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q3/5

763

QUESTION ID

S21-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$; for $x = y = 1$, $8 \neq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The number of terms in the expansion of $(x + y)^5$ is

A. 6

B. 5

C. 10

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q4/5

764

QUESTION ID

S21-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE n + 1 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(a - b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q5/5

765

QUESTION ID

S21-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$a^2 - 2ab + b^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 2

Build Pascal's Triangle by a rule • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Row 4 of Pascal's Triangle is 1, 4, 6, 4, 1. Write row 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q1/5

767

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1, 5, 10, 10, 5, 1

GRADING NOTE Form required: all six entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q1/5

Write row 7 of Pascal's Triangle (row 6 is 1, 6, 15, 20, 15, 6, 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q2/5

768

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1, 7, 21, 35, 35, 21, 7, 1

GRADING NOTE Form required: all entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q2/5

True or false?

Each inner entry of Pascal's Triangle is the sum of the two entries above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q3/5

769

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The rule of the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q3/5

The fourth entry of row 7 (counting from the first 1) is

A. 20

B. 7

C. 35

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q4/5

770

QUESTION ID

S21-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 35

GRADING NOTE 1, 7, 21, 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the sum of the entries in row 5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q5/5

771

QUESTION ID

S21-B2-Q5

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 3

Powers of products and of negatives • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(2a)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q1/5

773

QUESTION ID

S21-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$16a^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(-2y)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q2/5

774

QUESTION ID

S21-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$-8y^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q2/5

True or false?

$(-b)^4 = -b^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q3/5

775

QUESTION ID

S21-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An even power of $-b$ is positive: $(-b)^4 = b^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

3^4 equals

A. 27

B. 12

C. 64

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q4/5

776

QUESTION ID

S21-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 81

GRADING NOTE 3·3·3·3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(0.02)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q5/5

777

QUESTION ID

S21-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.0004

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 1

Find a specific term or coefficient with a condition • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Find the constant term of $(x + 1/x)^6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q1/5

779

QUESTION ID
S21-F1-Q1

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE Any equivalent form. Middle term C(6, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q1/5

The coefficient of x^2 in $(1 + kx)^5$ is 90. Find k ($k > 0$).

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q2/5

780

QUESTION ID
S21-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. $C(5, 2)k^2 = 10k^2 = 90$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q2/5

True or false?

The middle term of $(x + y)^8$ is $56x^4y^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q3/5

781

QUESTION ID

S21-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $C(8, 4) = 70$, so the middle term is $70x^4y^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q3/5

The term with x^4 in $(x^2 - 1)^5$ is

A. $-5x^4$

B. $-10x^4$

C. $10x^4$

D. $5x^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q4/5

782

QUESTION ID
S21-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $-10x^4$

GRADING NOTE $C(5, 2)(x^2)^2(-1)^3 = -10x^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q4/5

Find the coefficient of x in $(2x - 1)^6$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q5/5

783

QUESTION ID
S21-F1-Q5

TYPE
Short answer

CORRECT ANSWER
−12

GRADING NOTE Any equivalent form. $C(6, 1)\cdot 2\cdot(-1)^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 2

Use binomial coefficients to compute probabilities • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is tossed 5 times. Use C(5, k) to find P(exactly 3 heads).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q1/5

785

QUESTION ID
S21-F2-Q1

TYPE
Short answer

CORRECT ANSWER
10/32 = 5/16

GRADING NOTE Any equivalent form (0.3125).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner lands on red with probability 0.3. Find P(exactly 2 reds in 4 spins).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q2/5

786

QUESTION ID
S21-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.2646

GRADING NOTE Any equivalent form. $C(4, 2)(0.3)^2(0.7)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q2/5

True or false?

The probabilities of 0, 1, ..., n successes add to 1 because $(p + (1 - p))^n = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q3/5

787

QUESTION ID
S21-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Binomial Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For 6 fair coin tosses, P(exactly 3 heads) is

A. 6/64

B. 3/6

C. 15/64

D. 20/64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q4/5

788

QUESTION ID

S21-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20/64

GRADING NOTE C(6, 3) = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q4/5

Explain why C(n, k) counts the outcomes with exactly k successes in n trials.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q5/5

789

QUESTION ID
S21-F2-Q5

TYPE
Short answer

CORRECT ANSWER

Each outcome is a choice of which k of the n trials are successes; there are C(n, k) such choices.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q5**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q5/5

Rewrite rational expressions by polynomial division

MAIN	Algebra II main question	**HSA-APR.D.6**
BACKWARD 1	Write a division with a remainder as a mixed number	**4.NF.B.3**
BACKWARD 2	Multiply to check a division	**HSA-APR.A.1**
BACKWARD 3	Use the Remainder Theorem to check the remainder	**HSA-APR.B.2**
FORWARD 1	Divide with unknown coefficients and higher-degree divisors	**HSA-APR.D.6**
FORWARD 2	Use division to find slant asymptotes	**HSF-IF.C.7.d**

SET 22 • HSA-APR.D.6

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Rewrite $(x^3 + 2x^2 - 5x + 3)/(x - 1)$ in the form $q(x) + r(x)/b(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q1/5

792

QUESTION ID

S22-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x^2 + 3x - 2 + 1/(x - 1)$

GRADING NOTE REQUIRED WORK: Long division or synthetic division, shown.

Form required: q(x) + r/b(x). Synthetic division by 1: 1, 3, −2, remainder 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-M-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q1/5

Rewrite $(2x^3 - 3x^2 + 4x - 7)/(x^2 + 1)$ in the form $q(x) + r(x)/b(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q2/5

793

QUESTION ID

S22-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$2x - 3 + (2x - 4)/(x^2 + 1)$

GRADING NOTE REQUIRED WORK: Long division, shown.

Form required: $q(x) + r(x)/b(x)$ with deg $r < 2$. $(x^2 + 1)(2x - 3) = 2x^3 - 3x^2 + 2x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $(x^4 - 16)/(x - 2)$ in the form $q(x) + r(x)/b(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q3/5

794

QUESTION ID

S22-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$x^3 + 2x^2 + 4x + 8$ (remainder 0)

GRADING NOTE REQUIRED WORK: Long or synthetic division, using 0 coefficients for missing terms.

Form required: the quotient and the remainder 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-M-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q3/5

Rewrite $(6x^2 + x - 4)/(2x + 1)$ in the form $q(x) + r(x)/b(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q4/5

795

QUESTION ID

S22-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$3x - 1 - 3/(2x + 1)$

GRADING NOTE REQUIRED WORK: Long division, shown.

Form required: $q(x) + r/b(x)$. $(2x + 1)(3x - 1) = 6x^2 + x - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-M-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $(x^3 - 4x + 5)/(x + 3)$ in the form $q(x) + r(x)/b(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q5/5

796

QUESTION ID

S22-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$x^2 - 3x + 5 - 10/(x + 3)$

GRADING NOTE REQUIRED WORK: Synthetic or long division, using 0 for the missing x^2 term.

Form required: $q(x) + r/b(x)$. Synthetic division by -3: 1, -3, 5, remainder -10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 1

Write a division with a remainder as a mixed number • Grade 4 • Number & Operations–Fractions • 4.NF.B.3

Write 23/4 as a mixed number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q1/5

798

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5 3/4

GRADING NOTE Any equivalent form (5 + 3/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 47/5 in the form q + r/5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q2/5

799

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9 + 2/5

GRADING NOTE Form required: q + r/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

17/3 = 5 + 1/3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q3/5

800

QUESTION ID

S22-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 17 = 3 · 5 + 2, so 17/3 = 5 + 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q3/5

800